Road Risk Analysis Platform — Functional Module Documentation

Module Overview

Module	Objective	Main Role
AI Hazard Detection Module	Identify permanent road risks and infrastructure defects	Detect, locate, classify, and manage road hazards from recorded driving data
Predictive Cooperative Collision Prevention Module	Prevent potential collisions before they occur	Analyze vehicle interactions and provide proactive safety warnings
Community Reporting Module	Capture temporary road events reported by road users	Collect, validate, and distribute real-time road information from the community
Historical Risk Analysis Module	Identify dangerous areas based on historical patterns	Analyze accumulated driving and road data to identify risk zones and trends
Driver Behaviour Analytics Module	Analyze driving patterns and risky behaviours	Detect unsafe driving events and generate behavioural safety insights

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1. AI Hazard Detection Module

Sub-Module	Description	Functional Objective

Pothole Detection	Identification of damaged road surfaces and potholes	Detect recurring road defects that represent temporal risks
Traffic Sign Detection	Identification and classification of road signs	Monitor road signage availability, visibility, and compliancepermanent
Speed Bump Detection	Detection of speed control infrastructures	Identify speed bumps and their locations for driver awareness
Continuous Improvement and Retraining of AI Detection Models	Uses data from detections validated, rejected, or corrected by human reviewers to build and enrich training datasets for each detection model. Monitors historical results and model performance to identify when retraining is required.	Continuously improve the accuracy and reliability of all AI detection models by leveraging human validation feedback, accumulated data, and changes in model performance.
Parked/Roadside Vehicle Detection	Identification of vehicles creating possible road obstructions	Detect temporary physical obstacles affecting road safety
Hazard Location Management	Association of detected hazards with geographical locations	Ensure every detected risk can be located and communicated
Hazard Clustering	Grouping of nearby detections into road risk zones	Provide a consolidated view of repeated hazards in the same area
Detection Validation Workflow	Human review and approval process for detected hazards	Ensure reliability before hazards become available to users

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2. Predictive Cooperative Collision Prevention Module

Sub-Module	Description	Functional Objective
Telematics Data Collection	Periodic collection of telematics data from connected vehicles, including GPS position, speed, heading, and associated timestamps.	Provide the information required for real-time safety analysis
Live Vehicle Monitoring	Continuous monitoring of connected vehicles	Maintain awareness of active vehicles in the road environment
Road Context Analysis	Understanding vehicle position according to road structure	Differentiate meaningful interactions from simple geographic proximity
Trajectory Prediction	Estimation of future vehicle movements	Predict possible future conflicts between vehicles
Vehicle Conflict Detection	Identification of dangerous interactions between vehicles	Detect potential collision situations before they happen
Risk Evaluation Engine	Assessment of collision risk level	Determine the urgency of detected situations
Alert Management	Generation and management of driver warnings	Notify drivers when preventive action is required
Driver Safety Notification	Delivery of warnings through the user application	Help drivers react before a dangerous situation occurs
Road Intelligence Broker	Coordination of vehicle information and safety analysis	Manage relevant vehicle interactions within the road environment

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3. Community Reporting Module

Sub-Module	Description	Functional Objective

Road Event Reporting	User submission of road-related events	Allow drivers to contribute real-time road information
Traffic Condition Reporting	Reporting of congestion and traffic disruptions	Inform users about current traffic conditions
Police Presence Reporting	Reporting of checkpoints and enforcement activities	Warn drivers about temporary road controls
Accident Reporting	Reporting of road accidents	Provide immediate information about accident locations
Temporary Hazard Reporting	Reporting of temporary road dangers	Capture risks that cannot be detected permanently by AI
Road Closure Reporting	Reporting of blocked roads or unavailable routes	Help users avoid inaccessible roads
Weather Condition Reporting	Reporting of weather-related road risks	Communicate environmental conditions affecting safety
Roadside Assistance Requests	Reporting of assistance needs	Connect users with support services
Community Validation System	User confirmation or rejection of reports	Improve reliability through collective validation
Report Visibility Management	Control of report display according to geographical context	Show relevant information based on user location

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4. Historical Risk Analysis Module

Sub-Module	Description	Functional Objective

Driving Event Collection	Collection of historical driving-related events	Build a long-term understanding of road behaviour
Risk Data Aggregation	Grouping of historical safety information by geographical areas	Identify recurring risk patterns
Risk Score Calculation	Generation of safety indicators from historical data	Measure the risk level of locations and routes
Risk Zone Identification	Detection of areas with repeated safety problems	Identify dangerous locations requiring attention
Road Safety Heatmaps	Visualization of historical risk concentration	Provide a geographical view of safety trends
Historical Trend Analysis	Analysis of risk evolution over time	Understand whether road safety conditions improve or deteriorate
Route Safety Evaluation	Calculation of safety indicators for planned routes	Provide users with safer route recommendations
Risk Reporting Dashboard	Visualization of historical safety information	Support decision-making for authorities and operators

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5. Driver Behaviour Analytics Module

Sub-Module	Description	Functional Objective
Speeding Analysis	Detection of excessive speed events	Identify unsafe driving speed patterns
Harsh Braking Analysis	Detection of aggressive braking behaviour	Identify dangerous driving situations and possible road issues

Acceleration Analysis	Detection of sudden acceleration events	Identify risky driving behaviour
Cornering Analysis	Detection of unsafe turning behaviour	Identify dangerous driving patterns on curves and intersections
Driving Event Storage	Storage of behavioural events with location information	Maintain historical driving behaviour records
Behaviour Risk Scoring	Evaluation of driver behaviour severity	Generate safety indicators
Driver Behaviour Heatmaps	Visualization of risky driving concentration areas	Identify locations affected by unsafe behaviours
Driving Behaviour Statistics	Generation of behavioural reports and trends	Support fleet monitoring and safety improvement
High-Risk Zone Detection	Identification of areas with repeated unsafe behaviours	Reveal potential infrastructure or traffic problems

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Platform Integration View

Data Source	Related Module	Information Produced
Dashcam / Video Sources	AI Hazard Detection	Permanent road hazards
Connected Vehicles	Predictive Collision Prevention	Live vehicle safety information
Mobile Users	Community Reporting	Temporary road events
Vehicle Telemetry History	Historical Risk Analysis	Long-term road safety indicators
Driving Behaviour Data	Driver Behaviour Analytics	Driver and road behaviour insights

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User Application Information Layer

Information Type	Source Module	User Benefit
Road Hazards	AI Hazard Detection	Awareness of permanent risks ahead
Temporary Events	Community Reporting	Awareness of current road conditions
Collision Warnings	Predictive Collision Prevention	Prevention of immediate dangers
Route Safety Score	Historical Risk Analysis	Selection of safer routes
Driving Insights	Driver Behaviour Analytics	Improvement of driving behaviour

Details explanation of the modules

Module 1 — AI Hazard Detection Module

1. Purpose of the module

The AI Hazard Detection Module is responsible for identifying, locating, organizing, and preparing road hazards for validation using videos captured by dashcams or other supported video sources.

Its primary objective is to transform raw video footage into structured, geolocated, and reviewable road-hazard records that can be used by the Road Risk Analysis Platform.

The module answers four fundamental questions:

- What is on the road? Identify hazards such as potholes, traffic signs, speed bumps, and parked or roadside vehicles.
- Where is the hazard? Extract or reconstruct the geographical position associated with a detection.

- Is the detection reliable? Identify detections that need human review, correction, or rejection.
- How can detection quality improve over time? Use validation feedback and accumulated data to improve the AI models through controlled retraining.

The module therefore does more than run an AI model on video frames. It coordinates the detection pipeline, location extraction, result management, validation, evidence storage, clustering, and continuous improvement of the models.

Main processing principle

Video input

Uploaded video or supported video stream

AI hazard detection

Identify objects and road hazards in video frames

Geolocation and result processing

Associate detections with GPS coordinates, timestamps, and evidence

Validation and organization

Review uncertain detections, manage duplicates, and group nearby observations

Structured hazard records

Persist confirmed or reviewable results for use by the platform

Continuous AI improvement

Use appropriate human feedback and historical performance to improve future models

This diagram represents the logical workflow. In an implementation, some activities may run asynchronously or in parallel, and retraining is a separate process rather than a mandatory step for every video.

2. Scope of the module

The module covers four main hazard-detection categories and the supporting functions needed to make their results usable.

Component	Purpose	Expected result
Pothole Detection	Identify potholes and similar road-surface defects in video frames.	Pothole detections with associated evidence and location information where available.
Traffic Sign Detection	Identify and classify supported traffic signs. The project includes a model with approximately 78 classes.	Traffic-sign detections with the predicted class and associated evidence.
Speed Bump Detection	Identify speed bumps and similar road-calming features.	Speed-bump detections associated with the relevant video frames and location information.
Parked/Roadside Vehicle Detection	Identify vehicles stopped or positioned along the roadside when covered by the configured model.	Vehicle detections that can be reviewed and associated with the relevant context.
GPS Extraction and Reconstruction	Extract geolocation information from video overlays or available metadata and associate it with detections.	Coordinates and timestamps associated with observations, subject to data quality.

Detection Validation	Support the review, correction, acceptance, or rejection of AI-generated results.	More reliable hazard records and reference feedback for model improvement.
Hazard Clustering	Group detections that are geographically close or likely to represent the same hazard.	Better-organized hazard observations and reduced duplication.
Hazard Evidence and Data Storage	Preserve detection metadata, evidence references, location, and validation status.	Traceable records that can be reviewed and used by other platform functions.
Continuous Improvement and Retraining of AI Detection Models	Use appropriate human-review feedback and historical model performance to improve all detection models.	Versioned, evaluated models with measurable performance history.

Important distinction: geolocation is a cross-cutting capability, not a separate standalone hazard-management submodule. It supports detection, validation, reverse geocoding, clustering, and persistence throughout the workflow.

3. Detailed functioning of the module

3.1. Step 1 — Video acquisition and processing initialization

The process begins when the platform receives a video source for analysis.

The source may be:

- A video manually uploaded by a user.
- A video originating from a supported dashcam or streaming source.

- A recorded journey containing road scenes and, where available, GPS information embedded in a video overlay.

The platform initializes a processing task and associates it with the source. The task can then be processed in the background so that the user does not have to keep the application open while the video is analyzed.

The processing workflow can be represented as follows:

1. Receive the source

Receive the video or supported stream and identify its origin.

2. Initialize the processing task

Create a task record and track its source, status, progress, and errors where supported.

3. Prepare the video frames

Read the video and extract frames for analysis at the configured sampling frequency.

4. Run the detection models

Analyze the relevant frames using the models configured for each hazard category.

In the platform architecture previously discussed, this type of workload can be handled asynchronously using a processing queue and background workers. The Road Risk Analysis backend uses FastAPI, with Celery and Redis for background processing.

The exact sampling frequency, parallelism, and task-state implementation depend on the deployed configuration.

3.2. Step 2 — AI detection of road hazards

After the frames have been prepared, the configured AI models analyze them to identify relevant objects or road conditions.

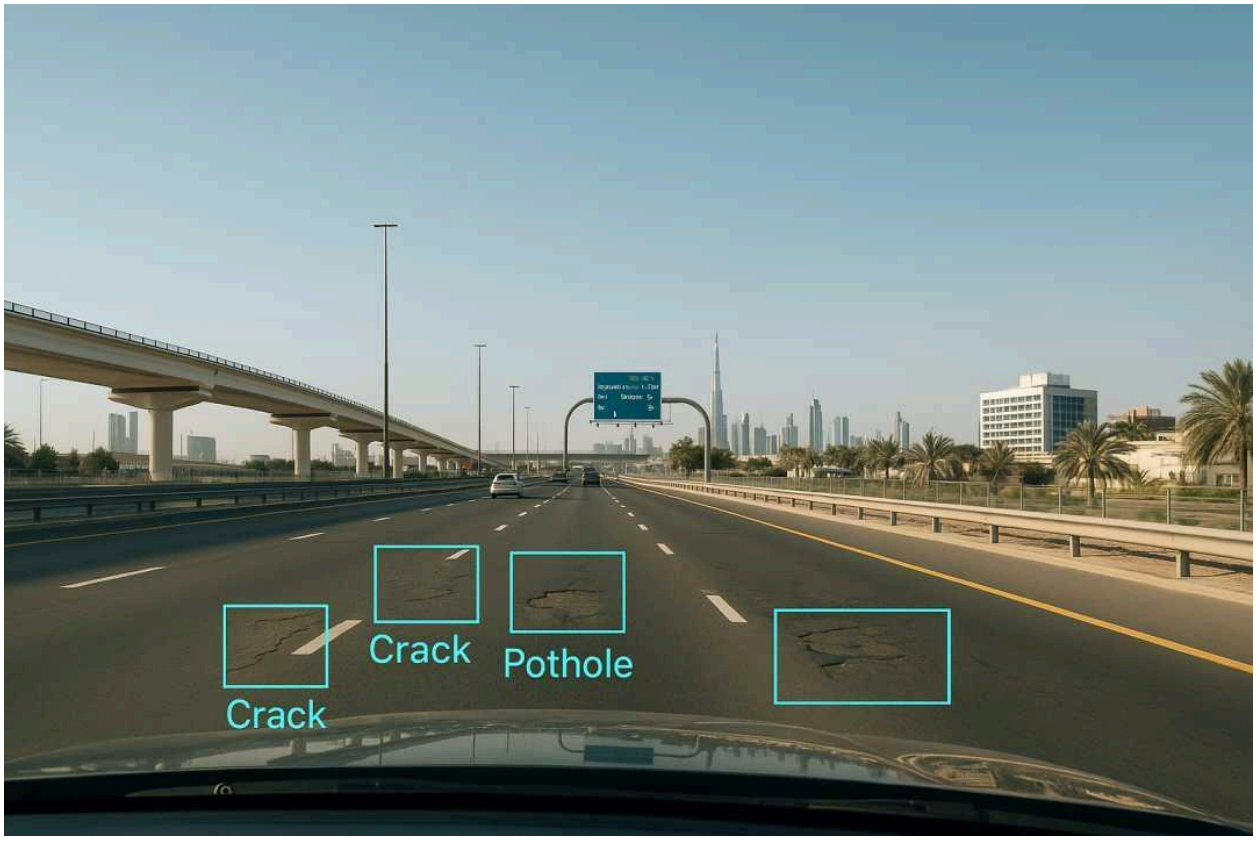
For example, a pothole model may identify a damaged road surface in a frame, while a traffic-sign model may recognize a sign and predict its class.

A typical detection result contains some or all of the following information:

Field	Meaning
Hazard category	Type of hazard identified, such as pothole, traffic sign, or speed bump.
Predicted class	Specific class predicted by the model, where applicable.
Confidence score	The model's confidence in its prediction. This is not, by itself, proof that the detection is correct.
Bounding box	The region of the frame in which the detected object appears.
Frame timestamp	The video time corresponding to the observation.
Source reference	Information identifying the originating video, stream, or processing task.
Evidence reference	Reference to an image or other retained evidence associated with the detection.

The bounding box is especially useful because it identifies the portion of the image that caused the model to produce a result. It also helps reviewers inspect detections without having to search manually through the entire video.

How the detection models differ



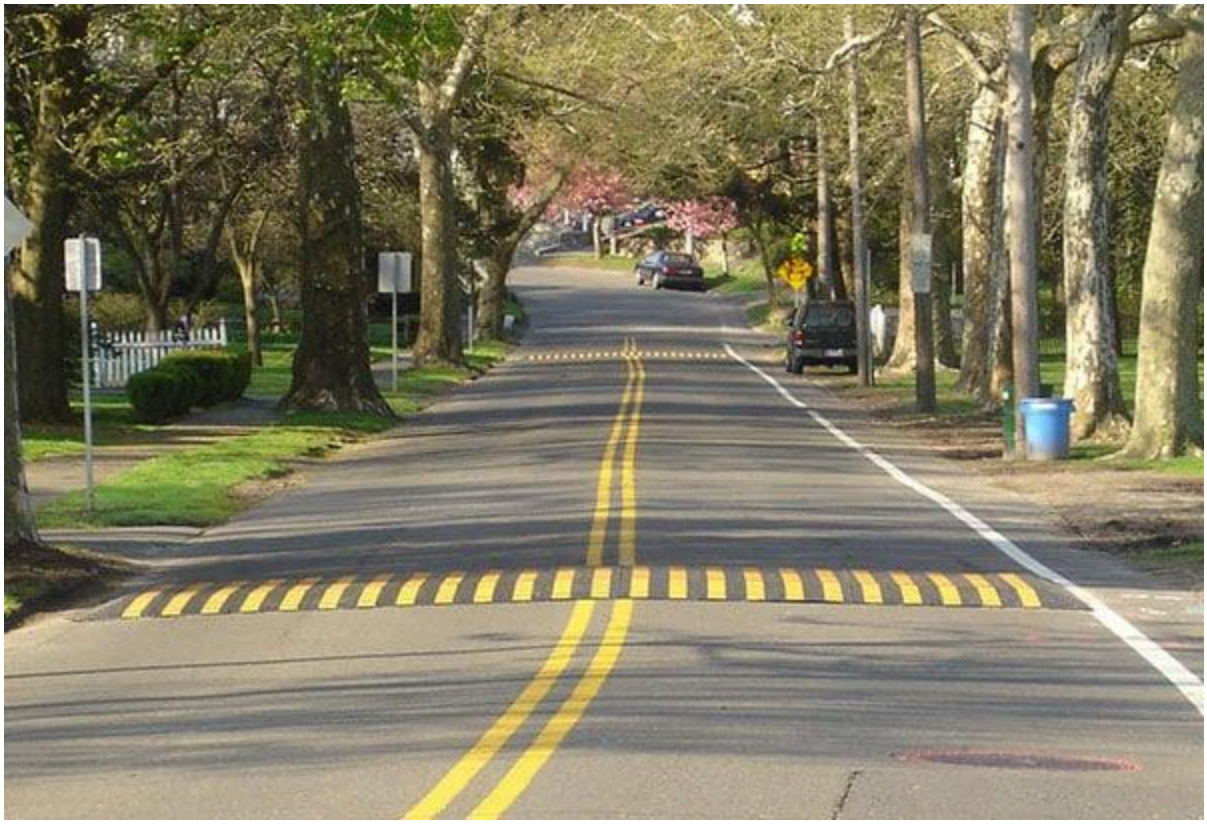
A. Pothole detection

The model identifies visible defects in the road surface. Its output can be used to locate the observation, retain evidence, and support subsequent validation. A separate severity assessment should not be assumed to be active in the production pipeline merely because a severity classifier exists.



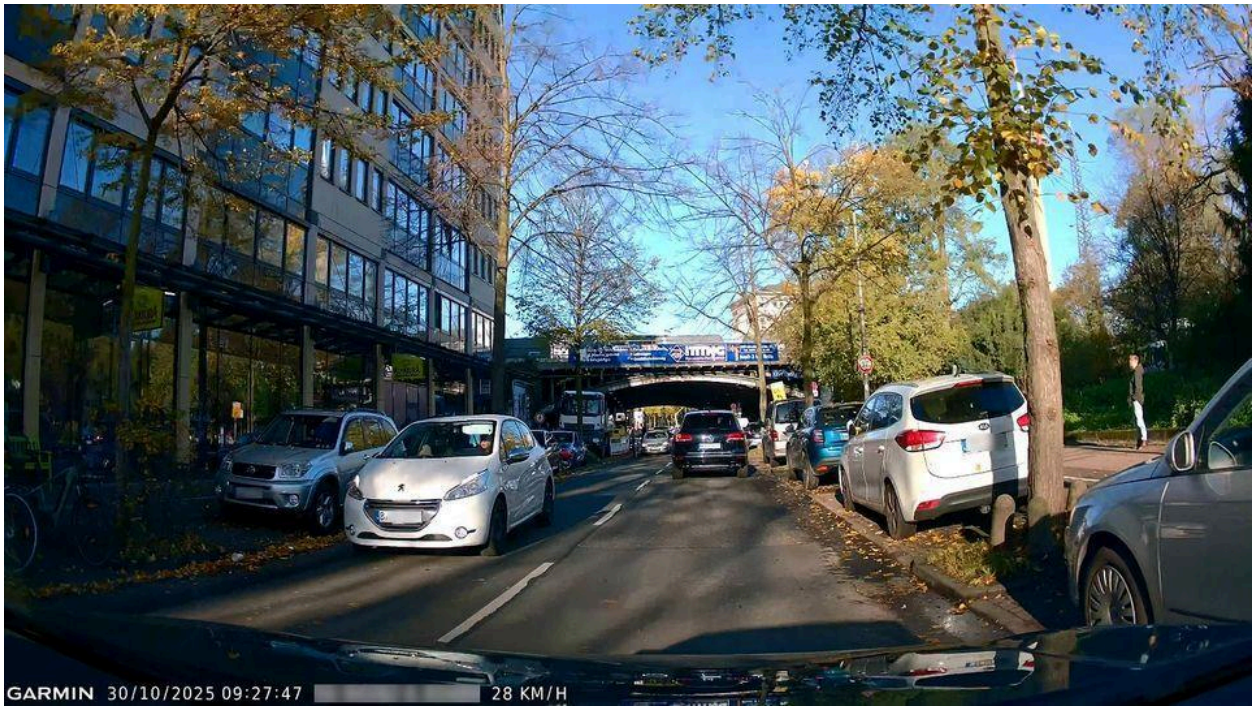
B. Traffic-sign detection

The model locates signs and predicts their classes. The project model covers approximately 78 classes. Its results can help build an inventory of observed signs and associate them with the locations where they were detected.



C. Speed-bump detection

The model identifies visible speed bumps and similar road-calming features. Their location and supporting evidence can then be attached to the corresponding detection records.



D. Parked or roadside vehicle detection

The relevant model identifies vehicles in the configured roadside-detection category. These observations can be reviewed and associated with their video context. Detecting a vehicle does not automatically establish that it is illegally parked or constitutes a road hazard; that conclusion requires appropriate rules or validation.

The detection models can operate on the same video, but each model has its own classes, confidence characteristics, and performance limitations. Their outputs should therefore remain identifiable by model and version.

3.3. Step 3 — Extracting the geographical position of a hazard

Detection answers the question, what was observed? Geolocation answers the question, where was it observed?

This is a critical stage because a hazard without a reliable location is difficult to use for mapping, road-risk analysis, or field intervention.

In the Road Risk Analysis Platform, GPS information can be obtained from available video metadata or from GPS information displayed as text over the video, such as a dashcam overlay.

When the GPS data is embedded visually in the video, the process may involve:

1. Selecting the relevant region of the frame containing the GPS overlay.
2. Preprocessing the cropped image to improve text readability.
3. Applying OCR, using a tool such as Tesseract, to extract the visible text.
4. Parsing the extracted text to identify coordinates, timestamps, or other supported GPS fields.
5. Checking the parsed values for validity and consistency.
6. Associating the resulting location with the corresponding detection and video timestamp.

The location should be derived from the video or its available metadata rather than assigned using a fixed road path.

Example

Suppose a dashcam frame contains a visible GPS overlay. The AI model identifies a pothole in that frame, while the OCR component extracts the corresponding coordinates.

The system can associate the following information:

Information	**Illustrative value**
Detected hazard	Pothole
Video timestamp	00:02:14
Latitude	4.0512
Longitude	9.7679
Evidence	Image of the frame containing the pothole

Location quality	Pending the applicable validation checks

The coordinates above are illustrative, not an actual detection from the platform.

If GPS extraction fails or the result is invalid, the system should not silently treat an uncertain position as an accurate one. Depending on the implemented rules, the detection can be flagged for review, retained without a confirmed location, or processed using an available reconstruction method.

Reverse geocoding

Once valid coordinates are available, a reverse-geocoding service can convert them into a human-readable location, such as a road name, locality, or nearby place.

It is important to distinguish between:

- GPS extraction: obtains coordinates from the video or its metadata.
- Reverse geocoding: converts coordinates into a readable geographic description.
- Hazard clustering: groups observations based on their spatial proximity or other applicable criteria.

These functions complement one another, but they solve different problems.

3.4. Step 4 — Detection post-processing and duplicate management

A single road hazard may appear in several consecutive frames. If every frame-level prediction is stored as an independent hazard, the platform could report the same pothole many times.

For this reason, the detection workflow needs to organize repeated observations.

A typical post-processing process may include:

- Filtering detections according to configured confidence or validity criteria.
- Retaining the source video and frame timestamp for traceability.
- Comparing detections from nearby frames or nearby coordinates.
- Grouping observations that are likely to refer to the same physical hazard.
- Selecting or retaining representative evidence according to the implementation rules.

For example, a pothole may appear in 15 frames while a vehicle passes it. Those 15 observations do not necessarily represent 15 separate potholes.

The system can use temporal and geographical information to identify that they may correspond to one physical hazard.

Clustering and duplicate removal are not identical. Duplicate management aims to avoid counting the same observation repeatedly. Clustering may group nearby hazards or observations for map visualization and further analysis, even when they are not necessarily the same physical object.

The criteria must be appropriate to the hazard type. Two traffic signs located close to one another, for instance, should not automatically be treated as duplicates.

3.5. Step 5 — Human validation of AI detections

AI predictions can be incorrect or incomplete. A model may misidentify an object, miss a real hazard, or produce a correct classification with an inaccurate location.

Human validation provides a mechanism for reviewing these results.

Depending on the configured workflow, detections can be:

- Accepted as valid.
- Corrected when the classification, location, or other information is wrong.
- Rejected when the prediction is not a valid detection.
- Flagged for further inspection when confidence or other quality criteria indicate uncertainty.

The platform has previously considered configurable review rules, including confidence-based review bands and detections occupying a very small portion of the frame. Such thresholds should be treated as configurable rules rather than universal values.

For example, a detection with a low confidence score might be sent for review instead of being automatically accepted. Similarly, a tiny bounding box could trigger review if the system's rules consider it potentially unreliable.

The confidence threshold must be configured according to the relevant model and validation objectives.

Why validation matters

Validation has two complementary purposes:

1. Operational reliability: it helps ensure that records presented as valid hazards have been checked according to the platform's applicable rules.
2. Model improvement: it provides feedback that can be used to identify weaknesses in the detection models and prepare better training data.

A rejected detection should not automatically become a negative training example without understanding why it was rejected. For example, a false-positive prediction and a genuine hazard that was incorrectly classified are different situations and should be handled differently during dataset preparation.

3.6. Step 6 — Store hazard records and supporting evidence

Once a detection has been processed, the platform needs to preserve the information required to understand, review, and use it later.

A structured hazard record may include:

Data group	Information to retain
Identification	Unique detection or hazard identifier.
Classification	Hazard category and predicted class.
Geographic information	Latitude, longitude, and readable location when available.
Temporal information	Video timestamp and relevant processing timestamps.

AI information	Model identifier or version, confidence score, and prediction details.
Evidence	Reference to the retained image or other supporting evidence.
Source information	Reference to the originating video, stream, vehicle, or processing task where applicable.
Validation	Review status, corrections, reviewer information, and reason for rejection or correction where supported.
Organization	Cluster or grouping reference, where applicable.

The platform uses PostgreSQL for structured data and MinIO for object storage in the previously described architecture. This supports the separation of structured hazard records from larger media objects.

The storage design should preserve traceability even if processed videos or temporary segments are removed. For example, a record can retain its source reference and relevant evidence while the original or intermediate video file follows a separate retention policy.

The exact database schema and retention rules depend on the deployed implementation.

4. Continuous improvement and retraining of all AI detection models

This submodule is particularly important because the platform is intended to improve its detection quality as more observations are collected and reviewed.

Its scope is not limited to pothole severity classification. It covers the AI detection models used for potholes, traffic signs, speed bumps, and parked or roadside vehicles.

4.1. Why continuous improvement is necessary

Road environments vary significantly. Lighting, weather, road surfaces, camera angles, image quality, and local road characteristics can affect model performance.

A model that performs well under one set of conditions may produce more false positives or miss hazards under different conditions.

Human-reviewed detections and accumulated historical results can help identify these weaknesses and provide better training examples.

4.2. How the improvement process works

1. Collect feedback

Collect detections that have been accepted, corrected, or rejected by reviewers, together with the relevant reasons and evidence where available.

2. Prepare datasets by hazard category

Organize suitable reference examples into datasets for the corresponding detection models.

3. Monitor model performance

Review historical results and available performance metrics to identify potential weaknesses or degradation.

4. Decide whether retraining is needed

Apply defined criteria to determine whether a new training cycle is justified.

5. Train and evaluate a candidate model

Train using the prepared dataset and evaluate the candidate against appropriate test data and acceptance criteria.

6. Manage deployment and versions

Promote the candidate to production only after it meets the required criteria, while preserving model-version and performance history.

This workflow describes the intended improvement lifecycle. It does not imply that every stage is already automated or that retraining occurs automatically after each validation.

4.3. Model performance and version traceability

The platform should maintain a history that makes it possible to determine which model produced a detection and how model performance changed over time.

Depending on the evaluation process, relevant metrics may include precision, recall, false-positive rate, and performance by hazard category.

For example:

- A high number of false-positive pothole detections may indicate that the model confuses shadows or road markings with road damage.
- Missed traffic signs may indicate difficulties with small objects, image blur, or certain sign types.
- A reduction in speed-bump detection performance may indicate that the model needs additional representative training examples.

These are examples of how performance monitoring can guide improvement, not claims that these specific problems have already been observed in production.

Keeping version history makes it possible to compare a candidate model with the currently deployed model, investigate changes in results, and identify which version generated a particular detection.

5. Relationship with the other Road Risk Analysis modules

Module 1 is an important source of road-environment information, but it is not responsible for every form of risk analysis in the platform.

Related module	Relationship with Module 1
Live Map and hazard visualization	Uses appropriately stored and geolocated hazard records to display road observations.
Community Reporting	Can complement AI-generated observations with reports submitted by users. A community report is not automatically an AI detection.
Historical Risk Analysis	Can use hazard-related data where the relevant data integration and analytical rules support it. It also has its own historical behavior-event inputs.
Driver Behaviour Analytics	Focuses on driver behavior events, such as speeding or harsh braking, rather than the visual identification of static road hazards.
Predictive Cooperative Collision Prevention	Focuses on vehicle telemetry, trajectory prediction, and potential vehicle conflicts. It complements Module 1 but uses a different detection and analysis process.

The distinction is important: Module 1 primarily identifies what road hazards are visible and where they were observed, while the predictive and behavioral modules analyze other dimensions of road safety.

6. What the module produces

At the end of the processing workflow, the platform should be able to manage structured records containing the hazard classification, associated geographic information, video timestamp, evidence references, processing provenance, and validation status where available.

These records provide the foundation for:

- Mapping and reviewing observed hazards.
- Organizing nearby observations and reducing duplication.
- Supporting subsequent road-risk analysis.
- Investigating AI predictions and their supporting evidence.
- Preparing reference datasets for improving detection models.

A detection should not be considered fully reliable merely because an AI model returned a result. Its confidence, location quality, validation status, and supporting evidence determine how it should be used downstream.

7. Important implementation distinctions

The following distinctions help avoid overstating the module's capabilities:

Capability	Interpretation
AI detection	Identifies potential hazards in video frames; model predictions can contain errors.
GPS extraction	Obtains coordinates from available video information; accuracy depends on the source and extraction quality.
Reverse geocoding	Converts coordinates into a readable geographic description; it does not independently verify the detected hazard.
Clustering	Organizes spatially or otherwise related observations; it does not prove that grouped detections are identical.

Human validation	Provides a mechanism to review and correct predictions; its effectiveness depends on the actual review workflow.
Severity assessment	Should be documented separately from basic pothole detection unless its integration into the active pipeline is confirmed.
Model retraining	Uses prepared data and evaluation criteria to improve models; collecting validation feedback alone does not guarantee that a new model has been trained or deployed.

Module 2 — Predictive Cooperative Collision Prevention Module

1. Purpose of the module

The Predictive Cooperative Collision Prevention Module is responsible for monitoring the positions and movements of connected vehicles in near real time, predicting potential conflicts between them, evaluating the associated collision risk, and delivering timely warnings to drivers.

Unlike Module 1, which identifies visible road hazards from video, Module 2 focuses on the interaction between moving vehicles and the possibility that their projected movements could lead to a collision.

Its primary objective is to detect dangerous situations early enough to give drivers an opportunity to react.

The module is built around a cooperative principle: participating vehicles periodically transmit their telemetry data to a central backend. The backend combines information from multiple vehicles, evaluates their relative movements in the road network, and makes relevant alerts available to the appropriate vehicles.

The system answers five fundamental questions:

- Where is each connected vehicle? Obtain its latest GPS position, speed, heading, and associated timestamp.
- Which vehicles could interact dangerously? Identify nearby vehicles whose road positions and projected movements may create a conflict.
- What could happen next? Estimate their future trajectories and determine whether they may converge.
- How serious is the situation? Evaluate factors such as distance, relative speed, time to collision, and applicable risk thresholds.
- Which driver should be warned? Create or update a relevant alert and make it available to the corresponding mobile application.

The module therefore goes beyond displaying vehicle positions on a map. Its purpose is to transform telemetry into forward-looking safety alerts.

Main processing principle

Connected vehicles

Mobile applications register vehicles and periodically transmit GPS telemetry

Telemetry ingestion

Receive vehicle position, speed, heading, and timestamp

Road context and vehicle state

Associate vehicles with road segments and directions; maintain their latest states

Trajectory prediction and conflict detection

Project movements, filter nearby vehicles, and identify potential conflicts

Risk evaluation and alert management

Evaluate distance, relative speed, time to collision, and risk thresholds

Driver safety notification

The mobile application retrieves relevant alerts and presents warnings to the driver

The workflow separates telemetry reception from risk computation so that a vehicle can transmit its latest state without waiting for the complete prediction and alert-processing pipeline to finish.

2. Scope of the module

The module consists of several functional components that work together to detect and communicate potential vehicle conflicts.

Component	Purpose	Expected result
Vehicle Registration and Identification	Register a participating vehicle or device and associate incoming telemetry with the correct identity.	A recognized vehicle that can participate in the cooperative safety system.
Telematics Data Collection	Receive periodic GPS position, speed, heading, and timestamp information from connected vehicles.	Structured telemetry updates available to the backend.
Live Vehicle Monitoring	Maintain and update the latest known state of each participating vehicle.	A current operational representation of connected vehicles.
Road Context Analysis	Associate vehicle positions with road-network geometry, road segments, and directions using available road-network information.	A road-aware representation of vehicle positions and movements.

Trajectory Prediction	Project the likely near-future movement of each vehicle using its current state and the applicable road context.	Estimated future positions or trajectories for conflict analysis.
Vehicle Conflict Detection	Compare relevant vehicles and identify situations where their projected movements may converge dangerously.	Potential vehicle conflicts requiring risk evaluation.
Risk Evaluation	Evaluate indicators such as distance, relative speed, time to collision, and configured thresholds.	A risk classification used to determine whether an alert should be generated or updated.
Alert Management	Create, update, deduplicate, and manage conflict alerts as the situation evolves.	Consistent alert records linked to the relevant vehicles and conflict.
Driver Safety Notification	Make relevant alerts available to the mobile application and present them through supported warning mechanisms.	Timely, understandable warnings for participating drivers.
Road Intelligence Broker	Coordinate telemetry-driven processing and the exchange of relevant vehicle-state, prediction, and alert information.	An organized backend workflow connecting the module's processing components.

The central backend performs the prediction and conflict analysis. The mobile application serves as the vehicle-facing client: it collects telemetry, transmits updates, retrieves alerts, and presents the resulting warnings.

3. Detailed functioning of the module

3.1. Step 1 — Vehicle registration and identification

Before a vehicle can participate in the cooperative collision-prevention process, it must be registered with the platform.

Registration establishes the association between the participating vehicle, its application or device, and the identity used by the backend to process telemetry.

The purpose is to ensure that every incoming update can be associated with the correct vehicle and that alerts can subsequently be returned to the appropriate client.

A typical registration workflow is:

1. The mobile application submits the vehicle registration information.
2. The backend validates the registration request.
3. The platform creates or retrieves the vehicle's identity.
4. The mobile application uses that identity when submitting telemetry updates.
5. The backend associates subsequent telemetry and conflict alerts with the registered vehicle.

Information	Purpose
Vehicle or device identifier	Identifies the source of telemetry.
Registration information	Associates the vehicle with its platform record.
Application or device association	Helps the backend determine which client is transmitting data and should retrieve alerts.
Registration status	Indicates whether the vehicle is recognized and permitted to participate.

The exact identity fields and authentication mechanisms depend on the deployed implementation. Registration alone should not be confused with continuous location tracking: tracking requires subsequent telemetry updates.

3.2. Step 2 — Telematics data collection

Once registered, the vehicle's mobile application periodically transmits its current telemetry to the backend.

The implemented workflow uses a periodic ping containing the vehicle's latitude, longitude, speed, and heading. The backend acknowledges receipt immediately with an HTTP `204 No Content` response, allowing the mobile application to continue its work without waiting for the complete risk analysis.

The important distinction is that receipt of telemetry and completion of collision analysis are separate operations.

The data collection process follows this sequence:

1. Collect the vehicle state

The mobile application obtains the available GPS position, speed, heading, and timestamp.

2. Submit a telemetry ping

The application sends the latest state to the backend through an HTTP request.

3. Acknowledge receipt

The backend returns HTTP 204, indicating that the request was successfully handled without a response body.

4. Continue background processing

Road matching, vehicle comparison, trajectory projection, and conflict evaluation proceed through the backend processing workflow.

Telemetry fields

Field	Description	Importance for collision prevention
Vehicle ID	Identifies the transmitting vehicle.	Allows telemetry to be associated with the correct vehicle state.
Latitude	Geographic north–south position.	Helps locate the vehicle on the road network.
Longitude	Geographic east–west position.	Helps locate the vehicle on the road network.
Speed	Current or most recently reported vehicle speed.	Helps estimate movement and relative closing speed.
Heading	Direction in which the vehicle is travelling.	Helps distinguish vehicles moving in the same direction from those approaching or crossing one another.
Timestamp	Time associated with the telemetry update.	Helps evaluate freshness and compare states reported at different times.

Other telemetry, such as acceleration, braking status, altitude, or vehicle type, may be useful extensions, but they should not be assumed to be mandatory inputs to the currently implemented conflict pipeline.

Why periodic updates matter: a position is only a snapshot. Collision prevention requires sufficiently fresh updates to estimate how vehicles are moving and whether their relationship is becoming dangerous.

If updates are delayed, missing, or inaccurate, the system's predictions may be less reliable. The backend should account for telemetry freshness when interpreting vehicle states and alerts.

3.3. Step 3 — Live vehicle monitoring and road matching

The backend uses incoming telemetry to maintain a representation of each participating vehicle's latest known state.

Raw GPS coordinates are useful, but they do not always indicate precisely which road segment a vehicle is using. Two vehicles can be geographically close while travelling on different roads, on opposite sides of a divided highway, or on roads that pass above or below one another.

To address this issue, the implemented workflow uses OSRM (Open Source Routing Machine) to associate vehicle positions with relevant road-network information.

This process is commonly called map matching or road snapping.

The backend can use the road network to derive or associate information such as:

- The relevant road segment or segment identifier.
- The direction of travel relative to the road.
- The road geometry and the vehicle's position along that geometry.
- Context needed to compare vehicles moving along the same or connected roads.

The road association is then used by the trajectory and conflict-analysis stages.

Why road context is essential

Consider two vehicles that are only a few metres apart in GPS coordinates. One might be travelling on a main road while the other is on a parallel service road. A simple distance-based system could mistakenly consider them to be on a collision course.

Road context helps distinguish this situation from one in which the vehicles are approaching each other on the same road.

However, OSRM provides road-network and routing information; it does not independently determine whether a collision will occur. Collision risk must be evaluated by the platform's own prediction and risk logic.

The quality of road matching also depends on the GPS accuracy, the available road-network data, and the suitability of the map-matching method for the situation.

3.4. Step 4 — Trajectory prediction

Once the backend has the latest vehicle states and relevant road context, it can estimate how the vehicles are likely to move over a short future period.

A trajectory represents a vehicle's estimated future movement rather than just its current position.

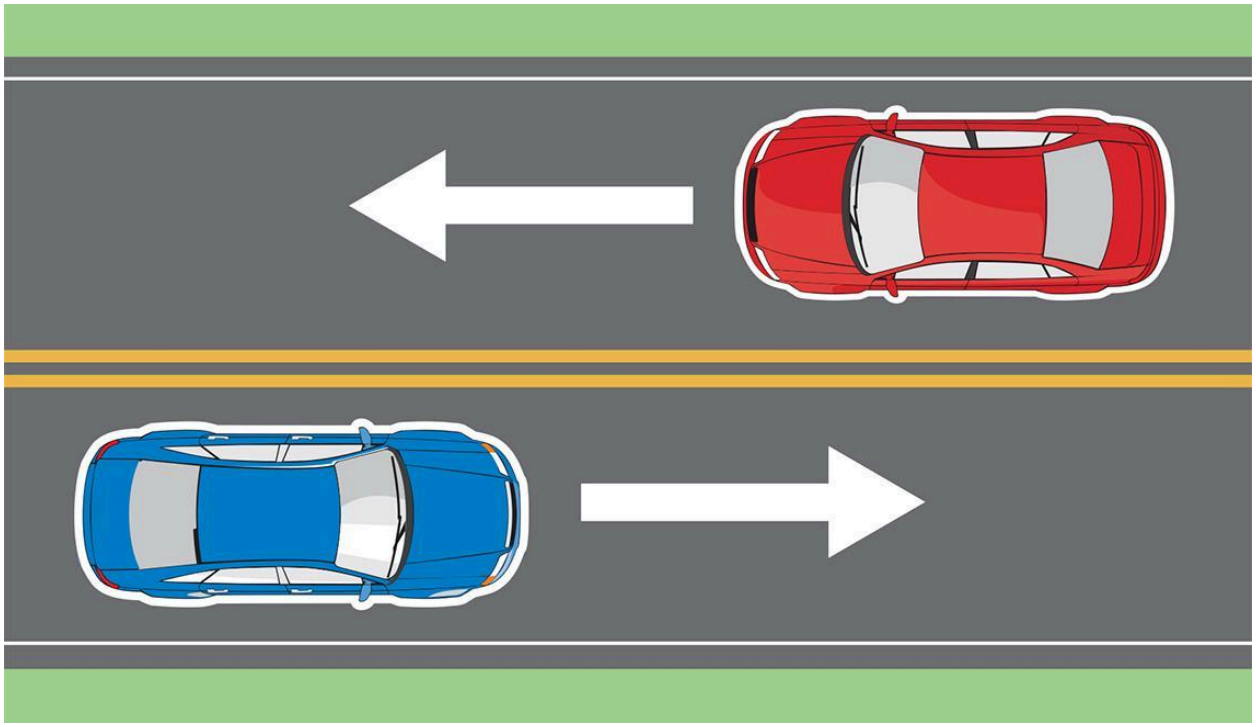
The prediction process considers the vehicle's location, speed, heading, and associated road geometry. The objective is to determine whether the vehicle is likely to remain on its current path, approach another vehicle, or reach an area where their projected movements may conflict.

For example:

- Two vehicles travelling in the same direction may be closing the gap between them.
- Two vehicles travelling in opposite directions may be approaching one another on the same road.
- Two vehicles on connected roads may be approaching a junction or another potential conflict point.

The prediction logic must take road direction and geometry into account instead of assuming that vehicles travel in straight lines indefinitely.

Illustrative example



aarp.org

Conceptual example: two vehicles travelling toward one another along the same road.

Vehicle A

60 km/h

Travelling east

Vehicle B

40 km/h

Travelling west

The backend evaluates their separation, opposing directions, estimated closing speed, and projected movement to determine whether a conflict may occur.

Trajectory prediction is an estimate, not a guarantee of future movement. A vehicle may brake, accelerate, turn, or change lanes after its last telemetry update. Consequently, predictions must be refreshed as new telemetry arrives.

The exact prediction horizon and mathematical trajectory model depend on the configured implementation. The purpose of this stage is to provide the conflict engine with a plausible representation of future vehicle movement.

3.5. Step 5 — Nearby-vehicle filtering and conflict detection

The platform does not need to compare every connected vehicle with every other vehicle at every update. It first narrows the search to vehicles that could plausibly interact.

In the currently implemented workflow, the backend filters nearby vehicles within an approximate 700-metre radius before performing the relevant conflict calculations.

This is a candidate-selection radius, not a claim that every vehicle within 700 metres is dangerous or that every possible conflict must fall within that distance.

For each relevant candidate pair, the backend evaluates information such as:

- The distance between the vehicles.
- Their respective speeds.
- Their relative movement and direction.
- Their road-segment association and road context.
- Their projected trajectories.
- Their estimated time to a possible conflict.

A vehicle can be geographically close without being on a collision course. Conversely, vehicles that are moving toward the same conflict point can require attention even if their current separation does not appear immediately dangerous.

Example scenarios

Scenario	What the backend evaluates	Possible interpretation

Same direction, similar speed	Separation and whether the gap is shrinking.	A conflict may be unlikely if separation remains stable, but the situation can change.
Same direction, one vehicle faster	Relative speed and time needed to close the gap.	A potential rear-end conflict may develop.
Opposing directions on the same road	Separation and closing speed along the road.	The vehicles may be approaching a head-on conflict.
Nearby but on separate roads	Road association, geometry, and projected movement.	The vehicles may be close geographically without posing a direct collision risk.
Approaching connected road segments	Trajectory convergence and estimated arrival at a shared conflict area.	A potential crossing conflict may require further evaluation if supported by the prediction logic.

These scenarios explain the underlying reasoning. The precise conflict types supported by the production engine depend on the implemented trajectory and risk rules.

3.6. Step 6 — Risk evaluation and time-to-collision calculation

After identifying a relevant vehicle pair, the backend evaluates whether the situation meets the configured conditions for creating or updating a conflict alert.

One important indicator is time to collision (TTC), which estimates how long it would take for two vehicles to reach a potential collision under the assumptions of the calculation.

For a simplified case in which two vehicles are approaching each other along the same line, TTC can be approximated as:

$$TTC = \frac{d}{v_{closing}}$$

Where:

- d is the current separation between the vehicles.
- $v_{closing}$ is their relative closing speed, expressed in compatible distance and time units.

If two vehicles are 100 metres apart and their combined closing speed is 20 metres per second, the simplified estimate is:

$$TTC = \frac{100}{20} = 5seconds$$

This is an illustrative calculation, not a measurement from the platform.

The formula only works when the assumed relative motion is meaningful and the closing speed is positive. It should not be applied blindly to vehicles travelling on different roads or whose projected trajectories do not intersect.

For more complex road situations, the system needs to evaluate the projected trajectories and the timing of their potential convergence.

Risk classification

The implemented workflow uses four risk-severity tiers based on configured thresholds. The system evaluates a conflict and creates or updates a `ConflictAlert` when the applicable conditions are met.

Conceptually, the four tiers represent increasing levels of urgency. The exact tier names, numeric TTC thresholds, and additional conditions must come from the deployed configuration; they should not be inferred from the number of tiers alone.

Risk evaluation can be summarized as follows:

1. Obtain the latest available states of the relevant vehicles.
2. Evaluate their distance and relative movement.

3. Estimate whether their projected trajectories may conflict.
4. Calculate or estimate TTC when applicable.
5. Apply the configured thresholds and conflict rules.
6. Determine whether an alert should be created, updated, maintained, or allowed to expire.

TTC is not a complete collision-probability model. It is an important indicator, but the actual risk depends on road geometry, trajectory assumptions, telemetry quality, and the conditions encoded in the risk engine.

3.7. Step 7 — Conflict alert creation and lifecycle management

When the risk engine identifies a situation that meets the configured alert criteria, the backend creates or updates a conflict-alert record.

The `ConflictAlert` represents a detected potential conflict between relevant vehicles. Rather than treating every telemetry ping as an entirely new event, the alert-management logic can update an existing alert as the situation changes.

This helps avoid overwhelming the mobile application with repeated notifications for the same evolving conflict.

A conflict alert may logically contain:

Information	Purpose
Alert identifier	Uniquely identifies the alert record.
Associated vehicle or vehicles	Identifies the participants in the potential conflict.
Conflict or risk classification	Indicates the risk tier assigned by the backend.

Relevant distance and timing indicators	Provides context for the conflict evaluation.
Creation and update timestamps	Records when the alert was generated and subsequently changed.
Alert status	Supports management of the alert as the situation evolves.
Retrieval information	Allows the appropriate client to obtain the relevant alert.

The actual database fields may differ from this conceptual list.

The lifecycle of an alert may involve:

- Creation: a new conflict satisfies the configured risk conditions.
- Update: new telemetry changes the estimated risk, distance, or other relevant indicators.
- Deduplication: repeated processing of the same evolving conflict should not unnecessarily create duplicate alerts.
- Resolution or expiration: the conflict no longer satisfies the active-alert criteria, or the alert becomes stale according to the applicable rules.

The exact resolution policy must be taken from the deployed implementation. An alert disappearing from a client's response does not, by itself, prove that a physical hazard has been resolved.

3.8. Step 8 — Alert retrieval by the mobile application

Once the backend has created or updated a conflict alert, the mobile application must retrieve it to inform the driver.

In the current implementation, the mobile application polls the backend for cached alerts. This means it periodically requests the alerts relevant to its registered vehicle rather than necessarily receiving every alert through a persistent real-time push connection.

The workflow is:

1. The backend processes incoming telemetry and updates the relevant conflict alerts.
2. The mobile application periodically requests the latest available alerts.
3. The backend returns the applicable cached alerts.
4. The application identifies new or updated alerts.
5. The application displays the warning and triggers the supported safety feedback.
6. Subsequent polling retrieves further changes as the situation evolves.

This approach separates alert computation from alert delivery. The backend can process telemetry independently, while the mobile client retrieves the results when it next checks for updates.

The distinction matters because the actual warning delay depends on several factors: telemetry frequency, background-processing time, polling interval, network conditions, and mobile operating-system restrictions.

3.9. Step 9 — Driver safety notification

The mobile application is the driver's interface with the collision-prevention system. Its role is to present backend-generated warnings clearly and promptly, without requiring the driver to interpret raw telemetry or risk calculations.

The currently implemented mobile experience includes:

- Radar-pulse animation: visually indicates a detected conflict or safety warning.
- Vibration: provides tactile feedback to draw the driver's attention.
- Alert display: presents the retrieved conflict information through the application's interface.

The notification should help the driver understand that a potential conflict has been detected and that attention is required.

The interface should also avoid creating unnecessary distractions. Repeated updates to the same alert should not trigger excessive vibrations or duplicate warnings unless the risk has changed enough to justify renewed attention.

Voice warnings and more advanced navigation-integrated instructions were considered as broader capabilities, but voice notification should not be described as an implemented feature based on the current workflow alone.

The mobile application is a warning aid, not a replacement for driver attention, safe driving practices, or the vehicle's own safety systems.

4. The role of the Road Intelligence Broker

The Road Intelligence Broker is the logical coordination layer that connects the incoming telemetry to the platform's vehicle-state, road-context, prediction, conflict, and alert-processing components.

Its purpose is to organize the information flow so that the system can evaluate interactions between vehicles instead of processing each vehicle's position in isolation.

At a functional level, this coordination involves:

1. Receiving or accessing incoming vehicle telemetry.
2. Making the latest relevant vehicle states available for comparison.
3. Coordinating road-context association and trajectory analysis.
4. Connecting potential conflicts to the risk-evaluation process.
5. Ensuring the resulting alerts are available to the appropriate downstream clients.

The broker should not be confused with the AI prediction algorithm itself. Its primary role is orchestration and information exchange; the trajectory-prediction and risk-evaluation components perform the analytical work.

Likewise, OSRM provides road-network information, while the platform's own logic determines whether vehicle movements constitute a potential collision risk.

The logical architecture can be summarized as follows:

Mobile applications

Vehicle registration, telemetry transmission, alert retrieval, and driver warnings

Road Intelligence Broker / backend orchestration

Coordinates vehicle-state updates and the processing workflow

Road context

OSRM association, road segments, and direction

Trajectory prediction

Estimated movement and potential convergence

Conflict detection and risk evaluation

Distance, relative movement, TTC, and four-tier risk classification

ConflictAlert management and retrieval

Create or update alerts for retrieval by the relevant mobile clients

This is a functional representation of the architecture, not a claim that every box is implemented as a separate microservice. Some responsibilities may be implemented within the same backend application or processing component.

5. Example of a complete collision-prevention scenario

Consider two connected vehicles approaching one another on the same road.



Vehicle A

The mobile application reports its GPS coordinates, speed, heading, and timestamp.



Vehicle B

A second registered vehicle transmits its own telemetry to the same backend.

1. Telemetry reception: the backend acknowledges each ping with HTTP 204 and continues processing the information in the background.
2. Road association: the system uses OSRM road information to determine whether the vehicles are associated with the same road segment or relevant connected road geometry.
3. Candidate filtering: the backend checks whether the vehicles fall within the approximate 700-metre candidate-search radius.
4. Trajectory analysis: their positions, headings, speeds, and road context are used to estimate their projected movements.
5. Conflict evaluation: the system evaluates separation, relative movement, estimated TTC where applicable, and the configured risk thresholds.
6. Alert management: if the applicable conditions are met, the backend creates or updates a `ConflictAlert` with the relevant risk tier.
7. Mobile retrieval: the relevant application obtains the alert through its polling mechanism.
8. Driver notification: the application presents the warning using its alert interface, radar-pulse animation, and vibration.

If the vehicles continue to approach one another, subsequent telemetry updates can change the estimated risk and update the existing alert. If their movements diverge or the conflict conditions cease to be satisfied, the alert can be handled according to the configured lifecycle rules.

This example illustrates the processing logic. It does not imply that the system can guarantee collision prevention or predict every possible manoeuvre.

6. Main inputs and outputs of the module

Category	Information
Primary inputs	Vehicle registration information and periodic telemetry containing latitude, longitude, speed, heading, and timestamp.
Supporting inputs	Road-network information, latest vehicle states, configured risk thresholds, and existing conflict alerts.
Intermediate results	Road-associated vehicle states, candidate vehicle pairs, projected trajectories, distance and relative-motion calculations, and estimated TTC where applicable.
Primary output	Created or updated `ConflictAlert` records classified according to the configured four-tier risk system.
Client-facing output	Alerts retrieved by the mobile application and presented through the supported warning interface.

Operational records	Vehicle identity, telemetry timestamps, alert creation or update information, and other supported diagnostic data.

7. Relationship with the other Road Risk Analysis modules

Module 2 focuses on interactions between connected vehicles. It complements the other modules but does not replace their responsibilities.

Related module	Relationship with Module 2
AI Hazard Detection — Module 1	Identifies visible road hazards from video. These observations may provide complementary road-safety information, but the current vehicle-conflict workflow should not be assumed to use every Module 1 detection automatically.
Community Reporting	Supplies user-submitted reports about road conditions or incidents. These can complement vehicle-conflict analysis if an appropriate integration is implemented.
Historical Risk Analysis	Analyzes historical risk patterns and road-related data. Historical risk can complement predictive safety analysis, but it is distinct from the live telemetry-based conflict process.
Driver Behaviour Analytics	Focuses on driving events such as speeding, harsh braking, and harsh cornering. These events can provide additional context when the corresponding integration is available.

Live Map	Can display vehicle positions or relevant alerts if those data are exposed to the map interface. Visualization alone does not perform collision prediction.

The key distinction is that Module 2 evaluates the evolving relationship between moving vehicles, while the other modules focus on visible hazards, community observations, historical patterns, or driver behaviour.

8. Important implementation distinctions and limitations

Capability	Current interpretation
Vehicle registration	Establishes the vehicle or device identity used by the telemetry workflow.
Periodic telemetry ping	Sends latitude, longitude, speed, and heading; the backend acknowledges the request with HTTP 204.
Background processing	Allows the backend to continue road matching, prediction, and conflict analysis after acknowledging the ping.
OSRM road matching	Associates vehicle positions with road-network information; it does not independently calculate collision risk.
Nearby-vehicle filtering	Uses an approximate 700-metre radius to select candidates for further analysis. This is not itself a risk threshold.

Trajectory prediction	Estimates future movement based on available telemetry and road context. Its accuracy depends on data quality and prediction assumptions.
Risk classification	Uses four configured tiers. The exact labels and numeric thresholds must be taken from the actual configuration.
Conflict alerts	Are created or updated by the backend and made available through cached-alert retrieval.
Mobile notifications	The implemented client includes alert presentation, radar-pulse animation, and vibration. Voice alerts should be treated as a separate capability unless confirmed in the deployed version.
Collision prevention	Provides warnings about potential conflicts; it does not control the vehicles or guarantee that collisions will be avoided.

9. Overall objective and expected result

The Predictive Cooperative Collision Prevention Module transforms periodic vehicle telemetry into actionable safety warnings by combining vehicle-state monitoring, road context, trajectory prediction, conflict detection, risk evaluation, and alert delivery.

Its value lies in moving beyond a static view of vehicle locations. Instead, it continuously evaluates whether the movements of connected vehicles may create dangerous situations and makes relevant warnings available to drivers.

The expected outcome is a backend-driven cooperative safety system that can identify potential vehicle conflicts early, manage alerts consistently as conditions change, and provide timely warnings through the participating mobile applications.

Module 3 — Community Reporting Module

1. Purpose of the module

The Community Reporting Module enables users of the Road Risk Analysis Platform to report road hazards, dangerous situations, and other road-related incidents they encounter in their daily journeys.

Its primary objective is to complement the platform's AI-based hazard detection capabilities with information collected directly from road users. While Module 1 identifies hazards from video footage, the Community Reporting Module allows people to report conditions that they observe in the field, including situations that may not yet have been detected by an AI model.

The module also supports community-based verification. Other users can interact with submitted reports by confirming that a reported hazard is present or indicating that it is no longer there. This feedback helps the platform assess the relevance of reports and manage their lifecycle.

The module answers five fundamental questions:

- What has been reported? Identify the hazard or incident category and the information supplied by the reporting user.
- Where is it located? Associate the report with GPS coordinates and, where available, a readable geographic location.
- Is the report supported by evidence? Preserve optional photographs, videos, or other submitted media.
- Do other users confirm the report? Collect community feedback to help assess whether the reported condition is still present.
- When should a report be cancelled? Apply the configured rules for removing or cancelling reports that are no longer considered valid.

The module therefore does more than provide a form for submitting road incidents. It establishes a workflow for collecting, locating, enriching, validating, and managing community-reported road information.

Main processing principle

Road user observes a hazard

A user identifies a road hazard or incident

Report submission

Submit the category, location, and optional media

Location processing and storage

Process GPS coordinates, obtain a readable location where possible, and store the report

Community feedback

Other users confirm the report or indicate that the hazard is no longer present

Report lifecycle management

Update the report's status according to the applicable rules and feedback

This is the logical workflow of the module. The exact moderation process, report statuses, and notification mechanisms depend on the implemented application.

2. Scope of the module

The Community Reporting Module consists of several functional capabilities that work together to manage reports from submission to their subsequent review and lifecycle management.

Component	Purpose	Expected result
Community Report Submission	Allow users to submit a road-related report by specifying its type, location, and optional supporting media.	A new report containing the information provided by the user.

Hazard and Incident Classification	Categorize reports according to the types of hazards or incidents supported by the platform.	A structured report that can be filtered and analyzed by category.
GPS Location Capture	Associate a report with the geographical coordinates of the reported situation.	A geographically identifiable report.
Reverse Geocoding	Convert coordinates into a readable geographic description when the service can resolve the location.	A more understandable location description, such as a locality or road name.
Media and Evidence Management	Preserve references to optional images, videos, or other submitted media.	Supporting evidence associated with the report.
Nearby Report Discovery	Find reports in the vicinity of a specified location.	A list of nearby reports that can help users understand local road conditions.
Route-Based Report Discovery	Retrieve relevant reports associated with a route or a specified journey.	Road-condition information relevant to the selected route, subject to the implemented query logic.
Community Confirmation	Allow users to indicate that a reported hazard is present.	Confirmation feedback associated with the report.

“Not There” Feedback	Allow users to indicate that a reported hazard is no longer present or cannot be found.	Negative confirmation feedback used by the report lifecycle rules.
Report Lifecycle Management	Apply the configured rules to maintain, update, or cancel reports as feedback is received.	Reports whose status reflects the applicable lifecycle rules.

These components support the core objective: collecting information from road users and making it more useful through location processing, evidence, community feedback, and lifecycle management.

3. Detailed functioning of the module

3.1. Step 1 — A road user identifies a hazard

The process begins when a user encounters a road condition or incident that they believe should be reported to the community.

Examples may include:

- A pothole or damaged road surface.
- A road obstruction.
- A dangerous road condition.
- Another supported road-related hazard or incident.

These examples illustrate the types of information a community reporting system can collect. The exact categories available to users depend on the application's configured report types.

The user initiates a report through the application's reporting interface. The purpose is to make the submission process straightforward enough for road users to contribute information while they are travelling or after observing a hazard.

The reporting interface collects the information necessary to identify and locate the reported situation.

3.2. Step 2 — Submit the community report

The report submission process allows the user to describe the situation and associate it with a location.

The core report information includes the report type, GPS coordinates, and optional media.

Information	Description	Purpose
Report identifier	Unique reference assigned to the report by the platform.	Allows the report to be retrieved, updated, and tracked.
Report type	Category selected by the user.	Identifies the kind of hazard or incident being reported.
GPS coordinates	Latitude and longitude associated with the reported situation.	Enables mapping and geographic searches.
Optional media	Photographs, videos, or other supported evidence supplied by the user.	Helps other users or reviewers understand the reported condition.
Submission timestamp	Time when the report was submitted, if recorded by the platform.	Supports report ordering and lifecycle management.
Reporting-user reference	Association with the reporting account or user, where supported.	Supports ownership, traceability, and applicable access rules.

Report status	The report's current lifecycle state, according to the implemented rules.	Indicates how the report should be treated by the application.

The table describes the logical data required or useful for report management. Not every field is necessarily mandatory, and the precise database schema may differ.

Submission sequence

1. Select the report category

The user identifies the type of road hazard or incident being reported.

2. Associate the location

The report receives GPS coordinates corresponding to the reported situation.

3. Add optional evidence

The user can attach supported media to illustrate the reported condition.

4. Submit the report

The application sends the report information to the backend for processing and storage.

Once submitted, the report becomes available to the platform according to the applicable visibility and status rules.

A key distinction is that submission does not automatically prove that a reported hazard exists. It means that a user has reported a situation that may require further community feedback or review.

3.3. Step 3 — Capture and process the geographical location

Geolocation is central to community reporting because a road hazard is useful only if other users can identify where it was observed.

The report is associated with GPS coordinates that can be used to position it on a map, retrieve nearby reports, and find reports relevant to a journey.

The geographical processing has two distinct responsibilities:

1. Capture or receive the coordinates associated with the report.
2. Convert those coordinates into a human-readable location when reverse geocoding is available.

Example

Suppose a road user reports a pothole in Douala.

The platform may store a record such as the following:

Field	Illustrative value
Report type	Pothole
Latitude	4.0512
Longitude	9.7679
Readable location	Obtained from reverse geocoding, if available
Supporting media	Optional photograph
Community feedback	Initially based on the feedback received so far

The coordinates are illustrative, not an actual report from the platform.

Reverse geocoding

GPS coordinates are precise geographic values, but they are not always easy for users to interpret. Reverse geocoding helps convert them into a readable description, such as a road name, neighbourhood, or locality.

The processing flow is:

GPS coordinates

Latitude and longitude associated with the report

Reverse-geocoding service

Resolve the coordinates against available geographic data

Readable location

Road, locality, or other geographic description when available

Reverse geocoding improves readability, but it does not verify that the reported hazard exists. Nor does a readable address necessarily identify the exact side of a road or lane on which the hazard is located.

If reverse geocoding fails, the coordinates can still be useful for geographic queries, provided that they are valid and retained.

3.4. Step 4 — Store the report and its supporting evidence

After submission and location processing, the report must be stored so that it can be retrieved, displayed, and updated as new feedback arrives.

The platform needs to preserve the relationship between the report, its coordinates, any attached media, and its community feedback.

Data group	Information to retain	Purpose
Report identity	Unique report identifier.	Enables reliable retrieval and updates.

Classification	Report category or incident type.	Supports filtering and category-based analysis.
Location	GPS coordinates and readable location, if available.	Supports mapping, nearby searches, and route-related queries.
Media	References to uploaded images or videos.	Allows users to inspect supporting evidence.
Community feedback	Confirmation and “not there” feedback associated with the report.	Supports community-based assessment of the report's relevance.
Lifecycle information	Current status and relevant timestamps, where supported.	Helps determine how the report should be displayed and managed.

Media and structured report information may be stored separately. For example, a database can retain the report record and its media reference while an object-storage service retains the actual photograph or video.

The exact storage technology and schema for this module should be confirmed against the deployed implementation rather than inferred solely from the report workflow.

3.5. Step 5 — Discover nearby reports

Once reports have been stored with valid coordinates, the platform can retrieve reports in the vicinity of a user's current or selected location.

The purpose is to make community information useful beyond the person who originally submitted it.

A nearby-report query generally works by:

1. Receiving the location around which reports are requested.
2. Searching stored report coordinates.
3. Applying the applicable geographic radius or spatial filter.
4. Returning the relevant reports and their available metadata.
5. Allowing the client to display those reports in a list or on a map.

For example, a driver approaching a particular area may want to know whether other users have reported potholes or road obstructions nearby.

The query can return matching reports, including their categories, locations, and available evidence, subject to the application's access and status rules.

Nearby does not necessarily mean relevant to the driver's road. A report may be geographically close but located on another street or on the opposite side of a divided road. Geographic proximity is therefore useful for initial discovery, while road or direction relevance may require additional filtering.

The exact search radius, sorting rules, and treatment of cancelled reports depend on the implemented query logic.

3.6. Step 6 — Discover reports along a route

The platform also supports route-related report discovery. This allows a user to retrieve community reports associated with a journey rather than searching only around a single point.

The purpose is to help users identify reported road conditions that may affect their intended route.

A typical route-based search follows this sequence:

1. Obtain the selected route or the information needed to define it.
2. Identify reports geographically relevant to that route.
3. Apply the implemented route-proximity or spatial filtering rules.
4. Return the matching reports for presentation to the user.

For example, a driver planning a journey through several neighbourhoods may want to see reports of potholes or obstructions along the way.

Route-related report discovery differs from nearby-report discovery:

Nearby-report search	Route-based report search
Starts from a particular location.	Starts from a route or journey.
Finds reports around a geographic point or area.	Finds reports relevant to the route's geographic path.
Useful for understanding local conditions.	Useful for identifying reported conditions that may affect a journey.

The existence of route-based report retrieval does not necessarily mean that the application automatically recalculates the route, proposes detours, or guarantees that every relevant report will be found. Those capabilities depend on additional routing and filtering functionality.

3.7. Step 7 — Community confirmation of a reported hazard

A report submitted by one user may be incomplete, inaccurate, or become outdated. The Community Reporting Module therefore allows other users to provide feedback about the reported situation.

The community confirmation mechanism gives users the opportunity to indicate that the reported hazard is present.

For example, if one user reports a pothole, another road user who encounters the same pothole can confirm that it exists. This adds supporting feedback to the original report.

The process can be described as follows:

A report is available

A user has submitted a road hazard report

Another user encounters the location

The user inspects the reported situation

Submit confirmation feedback

The user indicates whether the reported hazard is present

Update community feedback

The platform associates the feedback with the relevant report

Community confirmation is useful because it provides additional observations from people who may have visited the location after the original report was submitted.

However, the number of confirmations is not an absolute guarantee that the report is correct. Users may make mistakes, and several confirmations may reflect the same underlying misunderstanding. The feedback should therefore be interpreted according to the platform's validation rules.

3.8. Step 8 — “Not there” feedback

The module also supports a negative feedback action, referred to as “not there.”

This action allows a user to indicate that the reported hazard is no longer present or cannot be found at the specified location.

This distinction is important because road conditions can change over time. A pothole may be repaired, an obstruction may be removed, or a temporary incident may end.

The “not there” action contributes to the report's lifecycle management by providing information that may indicate the original report is no longer relevant.

The implemented rule is that a report is automatically cancelled after three “not there” votes.

The conceptual process is:

Active report

A road hazard has been reported

Users submit “not there” feedback

The platform records the negative feedback associated with the report

Evaluate the vote count

Check whether the configured threshold of three votes has been reached

Fewer than 3 votes

The automatic cancellation threshold has not yet been reached.

3 votes reached

The report is automatically cancelled according to the configured rule.

Important limitation: vote proximity validation

The current implementation does not validate whether a user submitting a confirmation or “not there” vote is physically close to the reported location.

This means the system cannot necessarily establish that a user personally visited the location before submitting feedback.

That limitation matters because the automatic cancellation rule relies on community input. If votes can be submitted from anywhere, the three-vote threshold alone does not prove that the hazard has disappeared.

The current behaviour should therefore be understood as a community-feedback rule, not a verified physical inspection.

Potential improvements include proximity checks based on a user's current GPS location, duplicate-vote prevention, vote eligibility rules, and audit records. These are recommendations for strengthening the system, not claims about capabilities already implemented.

3.9. Step 9 — Report lifecycle and status management

The report lifecycle determines how a report is treated after submission and as community feedback accumulates.

A useful conceptual distinction is between the report's existence, its current relevance, and the feedback received about it.

For example, a report can exist in the database even after it has been cancelled. Cancellation does not necessarily mean that the record must be permanently deleted.

Keeping historical records can support traceability, prevent old reports from being mistakenly recreated without context, and provide information about how community reports evolve over time.

Lifecycle event	Meaning	Expected handling
Report submitted	A user has reported a hazard or incident.	Store the report and make it available according to the application's rules.
Confirmation received	A user indicates that the hazard is present.	Record the confirmation against the report.
“Not there” feedback received	A user indicates that the hazard is no longer present or cannot be found.	Record the negative feedback and evaluate the applicable cancellation rule.
Three “not there” votes reached	The configured automatic cancellation threshold has been reached.	Automatically cancel the report.
Report cancelled	The report is no longer treated as active under the lifecycle rule.	Apply the configured display, retrieval, and historical-retention rules.

This table describes the known lifecycle behaviour and the associated functional expectations. Other statuses, such as pending moderation, rejected, resolved, or expired, should only be documented as existing states if they are confirmed in the application's implementation.

It is also important to distinguish between cancellation and deletion. The known rule specifies automatic cancellation; it does not establish that the corresponding database record or its supporting media are permanently deleted.

4. Community reporting data and integrity

Because this module depends on user-submitted information, data integrity is particularly important.

Unlike AI-generated detections, community reports originate from people and may contain incomplete information, inaccurate coordinates, outdated observations, or incorrect classifications.

The module should therefore maintain a clear relationship between the original report and all subsequent feedback.

4.1. Report and vote association

Each confirmation or “not there” vote needs to be associated with the report to which it applies.

This association allows the platform to determine which report received the feedback and evaluate the applicable rules.

The platform should also be able to distinguish separate users and prevent the same user from submitting repeated votes that artificially inflate the count, where such restrictions are part of the implemented rules.

The available implementation details establish the community feedback mechanism and the three-vote cancellation threshold, but they do not establish all the vote-integrity controls that may be present.

4.2. Location accuracy

The usefulness of a report depends partly on the accuracy of its coordinates.

If a report is associated with an inaccurate position, nearby-report searches and route-based discovery may return misleading results.

Location accuracy also matters when evaluating community feedback. A user may genuinely be near the submitted coordinates but still be unable to see the reported hazard because it is on another road, on the opposite side of a divided highway, or at a different point along the route.

Consequently, physical proximity is a useful signal, but it does not independently prove that a report has been correctly verified.

4.3. Evidence management

Optional photographs or videos can help users understand the report and provide context for subsequent feedback.

Evidence may show the road surface, the surrounding environment, or the nature of the reported obstruction. However, media should be interpreted alongside the location and report metadata.

For example, an image may show a real pothole but fail to establish exactly where it is located. Similarly, a photograph taken earlier may no longer reflect the current road condition.

The module's media-management capability should therefore be understood as supporting the report, rather than automatically proving that the reported condition is still present.

5. Relationship with the other Road Risk Analysis modules

The Community Reporting Module provides a human-generated source of road-safety information that complements the platform's other analytical capabilities.

Related module	Relationship with Module 3
AI Hazard Detection — Module 1	AI detection identifies potential hazards in video. Community reports provide a separate source of observations from road users. Combining the two can improve coverage if a suitable integration is implemented.
Predictive Cooperative Collision Prevention — Module 2	Module 2 evaluates potential conflicts between connected vehicles. Community reports can provide complementary road-condition information, but they do not automatically become inputs to collision prediction unless the relevant integration exists.

Historical Risk Analysis	Community reports may provide additional information about locations where road hazards have been reported. Their use in historical risk scoring depends on the relevant data pipeline and analytical rules.
Driver Behaviour Analytics	Focuses on driver behaviour, including events such as speeding and harsh braking. Community reporting primarily concerns road conditions and incidents rather than driver-behaviour events.
Live Map	Can display geographically located community reports, subject to the map's data integration and report-status filters.
Route and navigation functionality	Route-based report discovery can expose community-reported hazards relevant to a journey. Automatic rerouting or route optimization is a separate capability.

The distinction is that Module 3 focuses on collecting and managing reports from the community, while other modules identify hazards through AI, predict vehicle conflicts, or analyze historical and behavioural data.

6. Main inputs and outputs of the module

Category	**Information**
Primary inputs	User-submitted report category, GPS coordinates, and optional supporting media.

Geographic inputs	Report coordinates, the location around which nearby reports are requested, or a route used for route-based discovery.
Community feedback	Confirmations and “not there” votes associated with existing reports.
Processing operations	Location processing, reverse geocoding where available, report storage, geographic queries, feedback counting, and lifecycle-rule evaluation.
Primary outputs	Stored community reports with their categories, locations, and supporting media references.
Discovery outputs	Nearby reports and reports relevant to a requested route, according to the implemented filtering rules.
Lifecycle outputs	Updated community-feedback information and automatic cancellation when the three-“not there”-vote threshold is reached.
Supporting records	Report identifiers, feedback associations, and lifecycle information where recorded by the application.

7. Important implementation distinctions and limitations

Capability	**Current interpretation**

Community report submission	Users can submit reports with a type, GPS coordinates, and optional media.
Reverse geocoding	Converts coordinates into a readable location when available; it does not independently verify a report.
Nearby-report discovery	Retrieves geographically nearby reports according to the applicable query logic.
Route-based discovery	Retrieves reports relevant to a route according to the implemented filtering rules. It does not automatically imply detour recommendations.
Community confirmation	Allows users to indicate that a reported hazard is present. Confirmation is community feedback, not necessarily independent verification.
“Not there” feedback	Allows users to indicate that the reported hazard is no longer present or cannot be found.
Automatic cancellation	The report is automatically cancelled after three “not there” votes.
Vote proximity validation	The current implementation does not validate whether voters are physically near the reported location.

Evidence	Optional media provides context but does not independently prove the hazard's current status.
Report cancellation	Should not be equated with permanent deletion unless the implementation explicitly performs deletion.
Cross-module integration	Community reports can complement other modules, but automatic data exchange with those modules should not be assumed without a confirmed integration.

8. Recommended improvements

The following improvements could strengthen the reliability of the Community Reporting Module. They are recommendations, not a statement that these features already exist.

Improvement	Purpose	Expected benefit
Proximity validation for votes	Check whether the user's reported location is sufficiently close to the hazard when submitting feedback.	Reduce feedback from users who have not visited the location.
Duplicate-vote prevention	Prevent repeated votes from the same eligible user from artificially increasing a report's vote count.	Improve the integrity of community feedback.

Report freshness management	Consider the age of a report when determining whether it remains relevant.	Reduce the visibility of potentially outdated information.
Evidence-assisted review	Allow reviewers or eligible users to examine supporting media alongside report details.	Improve the quality of report assessment.
Clear lifecycle statuses	Distinguish active, cancelled, and other genuinely supported states.	Make report status easier to understand and manage.
Audit trail	Preserve relevant report changes and feedback events.	Improve traceability and help investigate disputed cancellations.
Cross-module integration	Define controlled ways to share relevant community information with other platform modules.	Improve road-safety analysis where the additional data is useful and reliable.

The most important improvement to prioritize is vote integrity, because the current automatic cancellation rule depends on three “not there” votes while the system does not validate voter proximity. Proximity validation and duplicate-vote prevention would help make that rule more reliable.

9. Overall objective and expected result

The Community Reporting Module transforms observations made by road users into structured, geographically located reports that can be discovered and evaluated by other members of the community.

Its value comes from combining four core capabilities:

1. Information collection: users submit road hazards or incidents, with a category, GPS location, and optional supporting media.

2. Geographic discovery: the platform supports finding reports near a location or along a route.
3. Community feedback: users can confirm a reported hazard or indicate that it is no longer present.
4. Lifecycle management: the platform records feedback and automatically cancels a report when three “not there” votes have been received.

The module creates a complementary source of road-safety information alongside AI-based detection and the platform's other analytical capabilities.

Module 4 — Historical Risk Analysis Module

1. Purpose of the module

The Historical Risk Analysis Module is responsible for analyzing accumulated historical driving-behaviour events to identify road locations that repeatedly present elevated safety risks.

Its primary objective is to transform historical driving data into geographically organized, measurable road-risk indicators that can support route planning, driver safety assessment, and road-infrastructure decision-making.

Unlike Module 2, which evaluates potential conflicts between connected vehicles using recent telemetry, Module 4 focuses on historical patterns. It examines events accumulated over time to identify locations where risky driving behaviours or other recorded road-safety events occur repeatedly.

The module answers five fundamental questions:

- Where have risky events occurred? Associate historical events with stable geographic areas.
- What types of events are recorded? Analyze speeding, hard braking, hard acceleration, harsh cornering, and other supported event categories.
- Which locations present higher historical risk? Aggregate events and their severities into normalized geographic risk scores.
- How safe is a point or route? Provide risk information for a specific location or journey.
- How can this information support decisions? Supply historical risk indicators for route assessment, driver-facing safety scores, and infrastructure analysis.

The module therefore does more than count historical events. It organizes them geographically, combines their severity into a risk measure, and exposes that information through point-based and route-based queries.

Main processing principle

Historical event data

Speeding, hard braking, hard acceleration, harsh cornering, and other supported events

Data ingestion and preparation

Import event records in batches or through supported manual imports; normalize and validate the data

Geographic aggregation

Associate events with stable geographic grid cells

Weighted risk calculation

Combine event severities into normalized risk scores between 0 and 1

Historical risk profiles

Make geographic risk information available for point and route queries

Safety assessment and decision support

Support route safety scores and analysis of persistent road-risk patterns

This diagram represents the logical processing workflow. Batch ingestion, geographic aggregation, risk calculation, and query handling may operate as separate jobs or services depending on the deployed architecture.

2. Scope of the module

The module consists of data-ingestion, geographic aggregation, risk-scoring, and query capabilities that turn historical events into useful safety indicators.

Component	Purpose	Expected result
Historical Event Ingestion	Import historical driving-behaviour events through batch processing or supported manual imports.	Historical event records available for analysis.
Event Normalization and Validation	Standardize event fields, check data quality, and prepare records for geographic and risk processing.	Consistent event records suitable for aggregation.
Geographic Event Association	Associate historical events with stable geographic grid cells.	Events organized by location independently of changing road-segment identifiers.
Event Categorization and Severity	Classify supported event types and apply the corresponding severity weights.	Structured event categories and weighted risk contributions.
Geographic Risk Aggregation	Combine events associated with each grid cell.	Geographic summaries of accumulated driving-behaviour events.
Historical Risk Score Calculation	Produce a normalized risk score between 0 and 1 for each applicable cell.	Comparable geographic risk indicators.

Historical Risk Profiles	Maintain summarized information about event patterns and risk levels for geographic areas.	Historical risk information that can be queried and interpreted.
Point and Radius Queries	Retrieve risk information around a specified geographic location.	Local risk information for a point or surrounding area.
Route Risk Analysis	Retrieve and combine relevant geographic risk information along a route.	A route-level safety assessment.
Driver Safety Score	Present a driver-facing safety score derived from the applicable risk-analysis process.	A summarized indicator of route safety or risk, according to the configured scoring rules.
Historical Data Integration	Incorporate data from available external sources where the integration is enabled.	Additional historical information for risk analysis.

Important status distinction: the Ymane historical-data import was previously identified as suspended. The module's ingestion capability should not be interpreted as proof that the scheduled Ymane import is currently running.

3. Detailed functioning of the module

3.1. Step 1 — Collect and import historical driving events

The process begins with the collection of historical event records from available data sources.

These records describe driving behaviours or road-safety events observed during previous journeys. They provide the historical evidence needed to identify locations where potentially dangerous situations occur repeatedly.

The supported event categories include:

- Speeding.
- Hard braking.
- Hard acceleration.
- Harsh cornering.
- Other supported driving-behaviour or road-safety events.

The module supports historical data ingestion through batch processing or manual imports. This is important when working with large datasets because historical records can be processed in groups instead of being submitted individually.

Example ingestion workflow

1. Obtain historical records

Receive historical driving events from an available source or an import file.

2. Import the data

Submit the records through the supported batch or manual import workflow.

3. Validate and normalize

Prepare event records for consistent geographic association and risk aggregation.

4. Make records available for analysis

Pass the prepared events to the geographic aggregation and risk-calculation stages.

A historical event may contain information such as:

Field	Purpose

Event type	Identifies the behaviour or road-safety event.
Event location	Provides the coordinates needed for geographic association.
Event timestamp	Indicates when the event occurred.
Event severity or measured value	Supports weighting and comparison when available.
Source reference	Identifies the source or import batch when recorded.
Vehicle or journey reference	Provides context for the event when available.

These fields describe useful data for the process; the precise schema and mandatory fields depend on the source format and implementation.

The role of Ymane data

Ymane was identified as a potential source of historical driving-event data for this module. However, its scheduled import is currently suspended.

Consequently, the module's historical-analysis workflow should be distinguished from the availability of that particular data feed. Risk analysis depends on the historical records actually imported and processed; it should not be assumed that new Ymane records are being ingested automatically while the scheduled import remains suspended.

3.2. Step 2 — Normalize and validate historical records

Historical records may originate from different sources or import batches. Before they can be combined, the module needs a consistent representation of the events.

Normalization prepares records for downstream processing. Validation helps identify records that are incomplete, malformed, duplicated, or unsuitable for geographic analysis.

For example, an event with a missing location may still be useful for some aggregate statistics, but it cannot reliably be assigned to a geographic cell without a suitable location-reconstruction or correction process.

Typical data-quality checks include:

- Ensuring that the event type is recognized.
- Checking that latitude and longitude contain valid values when location is required.
- Ensuring that timestamps use a consistent representation.
- Validating numeric measurements and severity values where applicable.
- Identifying duplicate records when the available identifiers or source data permit it.
- Preserving the relationship between the imported event and its source.

These checks help prevent invalid records from distorting geographic risk scores.

The exact rejection, correction, and deduplication rules should follow the implemented ingestion logic. Normalization should not silently invent missing event locations or convert uncertain data into confirmed facts.

3.3. Step 3 — Associate events with stable geographic grid cells

One of the module's most important design principles is that historical events are aggregated into stable geographic grid cells, independently of OSRM road-segment identifiers.

This allows the platform to build persistent geographic risk profiles without making the identity of a risk area dependent on a particular routing-engine segment ID.

Road-segment identifiers may change because of changes in road-network data, map versions, or segmentation. A stable geographic grid provides a consistent spatial framework for aggregating historical events across imports and analysis periods.

How the grid-based approach works

1. The module obtains the geographic coordinates of a historical event.
2. It determines which grid cell contains the event's location.

3. It associates the event with that cell.
4. It aggregates the event with other relevant records assigned to the same cell.
5. It calculates or updates the cell's historical risk information.

Historical events

Events with available geographic coordinates

Stable geographic grid

Each event is assigned to the appropriate spatial cell

Per-cell event aggregation

Combine the relevant historical events and their weighted severity

Geographic risk profile

A normalized risk score associated with the cell

Why stable cells matter

Imagine that several speeding and hard-braking events are recorded at approximately the same location over multiple weeks.

If those events are aggregated only by a road-segment ID, changes to the routing engine's segmentation could make it harder to compare the records consistently.

With stable geographic cells, the platform can continue to associate those observations with the same spatial area, provided that the grid definition and coordinate-assignment rules remain consistent.

This approach supports longitudinal analysis: the platform can compare accumulated events in an area over time and identify places where elevated risk persists.

Important distinction: a geographic cell is not necessarily a single road segment. A cell may contain multiple roads, intersections, or road directions. Its risk score represents the events assigned to that geographic area, not necessarily the risk of every road within it.

The grid's dimensions and resolution influence the result. Smaller cells provide more spatial detail but may contain fewer events; larger cells aggregate more data but can blur differences between nearby roads. The exact cell size is an implementation parameter and should not be assumed without the relevant configuration.

3.4. Step 4 — Categorize events and determine their risk contribution

Not every historical event contributes equally to the risk assessment.

For example, speeding, hard braking, and harsh cornering represent different driving behaviours. Their implications also depend on factors such as severity, location, and the configured risk model.

The module uses weighted event severities to calculate geographic risk scores. Each relevant event contributes according to the applicable weighting rules rather than every event necessarily counting as an identical unit.

Conceptually, the process involves:

1. Identifying the event category.
2. Determining the applicable severity or weight.
3. Associating the weighted event with its geographic cell.
4. Combining its contribution with those of other relevant events.
5. Applying the risk-calculation rules to produce a normalized score.

An illustrative weighting table might look like this:

Event category	Illustrative risk consideration
Speeding	The degree or severity of the speeding event, when the required measurements are available.
Hard braking	The severity and frequency of hard-braking events.

Hard acceleration	The severity and frequency of hard-acceleration events.
Harsh cornering	The severity and frequency of harsh-cornering events.
Accident or other supported event	The applicable event category and weighting, if such records are included in the processed dataset.

This table describes the types of considerations a weighted model can use. It does not establish the exact numeric weights or prove that every category is currently active in the same production scoring formula.

Why weighting is necessary

Consider two geographic cells:

- Cell A contains several low-severity events.
- Cell B contains fewer events, but some have much higher assigned severity.

A simple event count might rank Cell A as riskier because it contains more events. A weighted model can account for severity and potentially identify Cell B as the higher-risk area.

The objective is to produce a score that reflects more than event volume alone.

However, event weighting and normalization must be designed carefully. A high score should be interpreted according to the model's definition, data coverage, and calibration—not as a direct probability that an accident will occur.

3.5. Step 5 — Calculate the normalized geographic risk score

After the relevant events have been associated with geographic cells and their severity weights determined, the module calculates a risk score for each applicable cell.

The score is normalized to the range:

$$0 \leq R_{cell} \leq 1$$

Where R_{cell} represents the historical risk score associated with a geographic cell.

A value closer to 0 represents a lower score under the configured model, while a value closer to 1 represents a higher score. These values should be interpreted according to the actual scoring rules and calibration.

A score of 0.8 does not automatically mean an 80% probability of an accident. It is a normalized risk indicator, unless the model has been specifically calibrated and validated to represent a probability.

Illustrative interpretation

Example of normalized cell scores

Illustrative values only; these are not official risk-classification thresholds.

Cell A

0.18

Lower relative score

Cell B

0.52

Intermediate relative score

Cell C

0.86

Higher relative score

The calculation may take into account the weighted severity and accumulation of events assigned to the cell. Depending on the implemented model, it may also use additional factors or normalization rules.

The exact mathematical formula, weights, treatment of event frequency, and handling of cells with insufficient data must be taken from the actual scoring implementation. They should not be inferred solely from the fact that the output is normalized between 0 and 1.

What the score provides

The score allows geographic areas to be compared using a common numerical scale.

It can support:

- Identifying areas with elevated historical driving-event risk.
- Comparing the relative risk scores of cells.
- Retrieving a risk indicator for a location.
- Combining cell-level information into a route-level assessment.
- Tracking changes in geographic risk when historical data is updated.

The score remains dependent on the quality and coverage of the underlying records. An area with few recorded events may have limited evidence, rather than necessarily being safe.

3.6. Step 6 — Build historical risk profiles

A geographic risk profile summarizes the historical event information associated with a particular area.

In addition to the normalized risk score, a profile can contain aggregate information such as event counts, average or maximum recorded speeds, and risk classifications, where those metrics are supported by the available data and implemented profile.

The purpose is to make the underlying history more interpretable than a single score alone.

Profile information	Purpose
Geographic cell identifier	Identifies the stable spatial area being analyzed.

Historical event counts	Indicates the number of relevant events accumulated in the area.
Event categories	Shows which types of driving behaviours contribute to the observed risk.
Weighted severity information	Helps explain the relative contribution of events to the score.
Normalized risk score	Provides the cell-level risk indicator between 0 and 1.
Average or maximum speed metrics	Provides additional context where speed measurements are available and included in the profile.
Historical time information	Helps interpret when the events occurred, if the profile retains temporal aggregation.

The precise profile fields depend on the available dataset and implementation.

Why a profile is more useful than a score alone

Two areas can have similar normalized scores but different underlying event patterns.

For example, one area may have accumulated many speeding events, while another may have a smaller number of severe braking events. The overall scores might be similar even though the underlying driving patterns differ.

The historical profile helps provide context for interpreting the score and identifying which event categories contribute to the observed risk.

It is also important to distinguish between a historical profile and a live traffic state. A historical profile summarizes previously recorded events; it does not necessarily indicate the current traffic situation, an active obstruction, or a collision occurring at that moment.

3.7. Step 7 — Point and radius risk queries

The module supports queries for a specific geographic point and for an area around that point.

The purpose is to retrieve historical risk information relevant to a location without requiring the client to retrieve and process the entire historical dataset.

A point-based query can be used when an application needs a risk indicator for a specific location. A radius-based query can retrieve information from the surrounding area.

The logical workflow is:

1. Receive the requested coordinates.
2. Determine the relevant geographic cell or cells.
3. Retrieve the corresponding historical risk profiles.
4. Apply the requested geographic scope, such as a radius where supported.
5. Return the relevant scores and associated information.

Example

A user selects a point on the map and requests historical risk information.

The module identifies the relevant geographic cell and retrieves its risk profile. If the request specifies a radius, the module can retrieve the applicable information from the surrounding area according to the query implementation.

Query type	Primary purpose	Typical result
Point query	Assess historical risk at a specific location.	The applicable cell's risk score and available profile information.

Radius query	Assess historical risk around a location.	Relevant nearby cells or aggregated information for the requested area.

The exact response structure, maximum radius, cell-selection logic, and aggregation rules depend on the implemented API.

A point query should not be interpreted as an exact risk assessment for an entire road segment unless the spatial resolution and association rules support that interpretation.

3.8. Step 8 — Route risk analysis

One of the module's most important applications is the assessment of historical risk along a journey.

Rather than evaluating a single point, a route query allows the platform to retrieve historical risk information associated with the geographic path of a route.

The route can cross multiple geographic cells, each with its own accumulated event history and normalized risk score.

The general process is:

1. Obtain the route geometry or the information needed to identify the route.
2. Determine which geographic cells are relevant to the route.
3. Retrieve the applicable cell-level risk profiles.
4. Combine or summarize those profiles according to the configured route-scoring logic.
5. Return the route-level risk information to the requesting application.

Requested route

A route represented by its geographic path

Identify relevant grid cells

Find the geographic areas crossed by or relevant to the route

Retrieve historical risk profiles

Collect the normalized risk scores and relevant historical information

Calculate the route-level assessment

Apply the configured aggregation and scoring rules

Return the route safety score

Provide the historical risk indicator to the requesting application

Why route analysis matters

Suppose two routes connect the same origin and destination.

One route may pass through areas with relatively low historical risk scores, while another crosses several areas with elevated scores. The module can provide information that helps compare the historical risk characteristics of the two routes.

However, a route safety score is not necessarily a recommendation to choose the route with the lowest score. Travel time, road accessibility, route length, current conditions, and the scoring methodology may also matter.

The exact route-level formula must be taken from the deployed implementation. It should not be assumed to be a simple average of the cell scores: the implementation may use other aggregation rules, and the same risk score can have different implications depending on how much of the route passes through each area.

The module provides historical risk information to support route assessment; it does not automatically establish that every route query triggers rerouting or navigation changes.

3.9. Step 9 — Produce a driver-facing route safety score

The module supports a driver-facing safety score based on the historical risk information associated with the requested route or geographic context.

Its purpose is to summarize a potentially large collection of historical event records into a result that is easier for a user to understand.

The score can help users compare routes or understand whether a journey crosses areas with elevated historical risk.

The important distinction is between:

- Cell risk score: a normalized historical risk indicator for a geographic grid cell.
- Route risk score: an assessment derived from the relevant geographic information along a route.
- Driver safety score: a user-facing score whose meaning depends on the implemented scoring definition.

These terms should not automatically be treated as interchangeable. A cell score measures the risk indicator for an area, while a route score summarizes multiple areas. A driver-facing safety score may present or transform the route result according to the application's design.

The exact score direction, display scale, formula, and interpretation must follow the implementation. For example, it should be clearly established whether a higher displayed score means safer or riskier before the value is presented to drivers.

A safety score should also be accompanied by an appropriate explanation of its historical basis. It should not be presented as a guarantee of safety or as a prediction that an accident will occur.

4. Historical data integration and the Ymane import

Historical data integration is important because the quality of the risk analysis depends on the amount, consistency, and relevance of the available event records.

The module supports batch and manual ingestion, but one particular integration needs to be distinguished from the general capability: the scheduled import from Ymane is currently suspended.

This means that the module can process available historical records, but it should not be assumed that new Ymane events are being imported automatically.

Aspect	Functional role	Current status

Batch import	Process historical event records in groups.	Supported by the module's ingestion workflow.
Manual import	Allow historical records to be supplied through the supported import process.	Supported.
Historical event processing	Normalize and associate available records with geographic cells.	Part of the historical risk workflow.
Ymane scheduled import	Periodically bring historical event data from Ymane into the platform.	Suspended.
Risk-profile updates	Update geographic risk information using records that have been successfully processed.	Depends on the data available and the configured processing workflow.

The distinction is operationally important: an ingestion capability does not guarantee that every intended source is currently supplying data.

If the Ymane integration is reactivated, its imported records would need to pass through the applicable data-quality, geographic-association, and risk-calculation stages before contributing to updated risk profiles.

5. Relationship with the other Road Risk Analysis modules

Module 4 complements the platform's real-time, AI-based, and community-driven capabilities by providing a historical perspective on road safety.

Related module	Relationship with Module 4
AI Hazard Detection — Module 1	Identifies visible road hazards from video. Historical risk analysis focuses on accumulated events and their geographic distribution. The two sources may complement each other where an appropriate integration exists.
Predictive Cooperative Collision Prevention — Module 2	Evaluates potential conflicts between connected vehicles using recent telemetry. Module 4 identifies longer-term patterns from historical records rather than predicting an immediate vehicle-to-vehicle conflict.
Community Reporting — Module 3	Collects user-submitted reports about road hazards and incidents. Those reports may provide additional historical context if they are integrated into the relevant risk dataset and scoring process.
Driver Behaviour Analytics — Module 5	Focuses on analyzing driver behaviour events. Module 4 uses accumulated historical events to establish geographic risk profiles and support point- or route-level assessments. The exact data-sharing relationship depends on the implemented integration.
Live Map	Can visualize geographic risk information if the relevant profiles and scores are exposed to the map interface. The map itself does not calculate historical risk.

The key distinction is that Module 4 focuses on where historical risk has accumulated and how that information can support geographic and route-level safety assessment.

6. Main inputs and outputs of the module

Category	Information
Primary inputs	Historical driving-event records, including speeding, hard braking, hard acceleration, harsh cornering, and other supported event categories.
Geographic inputs	Event coordinates and geographic grid definitions.
Supporting information	Event severity or measurements, timestamps, and source references where available.
Data-processing operations	Batch or manual ingestion, normalization, validation, geographic association, event aggregation, and weighted risk calculation.
Geographic aggregation output	Stable geographic cells containing aggregated historical event information.
Primary analytical output	Normalized cell-level risk scores ranging from 0 to 1.
Historical profile output	Geographic summaries and relevant event metrics where supported.
Query outputs	Point-based and radius-based risk information.

Route-analysis output	A route-level historical risk assessment based on the relevant geographic profiles.
Driver-facing output	A safety score or risk indicator according to the implemented scoring rules.

7. Important implementation distinctions and limitations

Capability	Interpretation
Historical event ingestion	Supports batch and manual imports; does not imply that every external source is active.
Ymane integration	The scheduled import is currently suspended.
Stable geographic grid	Provides a consistent spatial aggregation framework independent of OSRM road-segment IDs.
Weighted event severity	Allows events to contribute differently to risk according to the applicable weights.
Normalized risk score	Produces a value between 0 and 1; this is not automatically an accident probability.

Geographic risk profile	Summarizes accumulated historical events for a geographic area; it is not necessarily a live traffic state.
Point and radius queries	Retrieve relevant geographic risk information according to the implemented spatial rules.
Route risk analysis	Combines information relevant to a route according to the configured aggregation method.
Driver safety score	Must be interpreted according to its actual definition, scale, and scoring direction.
Data coverage	A low score or absence of recorded events does not necessarily mean that an area is safe; the underlying data may be incomplete.
Infrastructure decisions	Historical profiles can support investigations and prioritization, but should be interpreted alongside data quality and other relevant evidence.

8. Recommended improvements

The following are potential improvements to strengthen the reliability and usefulness of the module; they are not claims that these features are already implemented.

Improvement	**Purpose**	**Expected benefit**

Reactivate and monitor the Ymane import	Restore the scheduled ingestion workflow when operationally appropriate.	Improve the availability of historical records from that source.
Track data freshness and coverage	Record the period covered by the data and identify geographic areas with limited observations.	Reduce the risk of misinterpreting incomplete data as low risk.
Version risk-scoring parameters	Keep a record of the weights and normalization rules used for each calculation.	Improve reproducibility and explain changes in scores.
Explain route scores	Provide relevant information about the cells or event categories contributing to a route assessment.	Make route risk results more interpretable to users.
Validate score calibration	Compare the score's behaviour against suitable historical evidence and evaluation criteria.	Improve the reliability of the risk indicator.
Track historical changes	Compare geographic risk profiles across defined time periods.	Help identify persistent patterns and changes in observed risk.

One important priority is to make risk scores interpretable in the context of data coverage. An area with a low score and extensive historical observations should not necessarily be treated the same way as an area with a low score but very few recorded events.

9. Overall objective and expected result

The Historical Risk Analysis Module transforms accumulated driving-behaviour events into geographically organized risk information.

Its main value comes from combining:

1. Historical data ingestion to make available driving-event records usable for analysis.
2. Stable geographic aggregation to associate events with consistent spatial cells.
3. Weighted risk calculation to account for the relative severity of accumulated events.
4. Normalized geographic risk scores to provide a common scale for comparing areas.
5. Point and route queries to make historical risk information available to applications.
6. Driver-facing safety assessment to summarize route-related historical risk in a more accessible form.

The expected result is a historical risk layer that helps the platform identify areas where risky driving events have accumulated and provide useful information for route assessment and road-safety decision-making.

In summary: Module 4 collects and processes historical driving events, assigns them to stable geographic grid cells, combines their weighted severities into normalized risk scores between 0 and 1, and exposes the resulting risk profiles through point, radius, and route queries. It provides a historical perspective that complements the platform's real-time collision-prevention, AI hazard-detection, and community-reporting capabilities.

Module 4 — Historical Risk Analysis Module

1. Purpose of the module

The Historical Risk Analysis Module is responsible for analyzing accumulated historical driving-behaviour events to identify road locations that repeatedly present elevated safety risks.

Its primary objective is to transform historical driving data into geographically organized, measurable road-risk indicators that can support route planning, driver safety assessment, and road-infrastructure decision-making.

Unlike Module 2, which evaluates potential conflicts between connected vehicles using recent telemetry, Module 4 focuses on historical patterns. It examines events accumulated over time to identify locations where risky driving behaviours or other recorded road-safety events occur repeatedly.

The module answers five fundamental questions:

- Where have risky events occurred? Associate historical events with stable geographic areas.
- What types of events are recorded? Analyze speeding, hard braking, hard acceleration, harsh cornering, and other supported event categories.
- Which locations present higher historical risk? Aggregate events and their severities into normalized geographic risk scores.
- How safe is a point or route? Provide risk information for a specific location or journey.
- How can this information support decisions? Supply historical risk indicators for route assessment, driver-facing safety scores, and infrastructure analysis.

The module therefore does more than count historical events. It organizes them geographically, combines their severity into a risk measure, and exposes that information through point-based and route-based queries.

Main processing principle

Historical event data

Speeding, hard braking, hard acceleration, harsh cornering, and other supported events

Data ingestion and preparation

Import event records in batches or through supported manual imports; normalize and validate the data

Geographic aggregation

Associate events with stable geographic grid cells

Weighted risk calculation

Combine event severities into normalized risk scores between 0 and 1

Historical risk profiles

Make geographic risk information available for point and route queries

Safety assessment and decision support

Support route safety scores and analysis of persistent road-risk patterns

This diagram represents the logical processing workflow. Batch ingestion, geographic aggregation, risk calculation, and query handling may operate as separate jobs or services depending on the deployed architecture.

2. Scope of the module

The module consists of data-ingestion, geographic aggregation, risk-scoring, and query capabilities that turn historical events into useful safety indicators.

Component	Purpose	Expected result
Historical Event Ingestion	Import historical driving-behaviour events through batch processing or supported manual imports.	Historical event records available for analysis.
Event Normalization and Validation	Standardize event fields, check data quality, and prepare records for geographic and risk processing.	Consistent event records suitable for aggregation.
Geographic Event Association	Associate historical events with stable geographic grid cells.	Events organized by location independently of changing road-segment identifiers.

Event Categorization and Severity	Classify supported event types and apply the corresponding severity weights.	Structured event categories and weighted risk contributions.
Geographic Risk Aggregation	Combine events associated with each grid cell.	Geographic summaries of accumulated driving-behaviour events.
Historical Risk Score Calculation	Produce a normalized risk score between 0 and 1 for each applicable cell.	Comparable geographic risk indicators.
Historical Risk Profiles	Maintain summarized information about event patterns and risk levels for geographic areas.	Historical risk information that can be queried and interpreted.
Point and Radius Queries	Retrieve risk information around a specified geographic location.	Local risk information for a point or surrounding area.
Route Risk Analysis	Retrieve and combine relevant geographic risk information along a route.	A route-level safety assessment.
Driver Safety Score	Present a driver-facing safety score derived from the applicable risk-analysis process.	A summarized indicator of route safety or risk, according to the configured scoring rules.

Historical Data Integration	Incorporate data from available external sources where the integration is enabled.	Additional historical information for risk analysis.

Important status distinction: the Ymane historical-data import was previously identified as suspended. The module's ingestion capability should not be interpreted as proof that the scheduled Ymane import is currently running.

3. Detailed functioning of the module

3.1. Step 1 — Collect and import historical driving events

The process begins with the collection of historical event records from available data sources.

These records describe driving behaviours or road-safety events observed during previous journeys. They provide the historical evidence needed to identify locations where potentially dangerous situations occur repeatedly.

The supported event categories include:

- Speeding.
- Hard braking.
- Hard acceleration.
- Harsh cornering.
- Other supported driving-behaviour or road-safety events.

The module supports historical data ingestion through batch processing or manual imports. This is important when working with large datasets because historical records can be processed in groups instead of being submitted individually.

Example ingestion workflow

1. Obtain historical records

Receive historical driving events from an available source or an import file.

2. Import the data

Submit the records through the supported batch or manual import workflow.

3. Validate and normalize

Prepare event records for consistent geographic association and risk aggregation.

4. Make records available for analysis

Pass the prepared events to the geographic aggregation and risk-calculation stages.

A historical event may contain information such as:

Field	Purpose
Event type	Identifies the behaviour or road-safety event.
Event location	Provides the coordinates needed for geographic association.
Event timestamp	Indicates when the event occurred.
Event severity or measured value	Supports weighting and comparison when available.
Source reference	Identifies the source or import batch when recorded.
Vehicle or journey reference	Provides context for the event when available.

These fields describe useful data for the process; the precise schema and mandatory fields depend on the source format and implementation.

The role of Ymane data

Ymane was identified as a potential source of historical driving-event data for this module. However, its scheduled import is currently suspended.

Consequently, the module's historical-analysis workflow should be distinguished from the availability of that particular data feed. Risk analysis depends on the historical records actually imported and processed; it should not be assumed that new Ymane records are being ingested automatically while the scheduled import remains suspended.

3.2. Step 2 — Normalize and validate historical records

Historical records may originate from different sources or import batches. Before they can be combined, the module needs a consistent representation of the events.

Normalization prepares records for downstream processing. Validation helps identify records that are incomplete, malformed, duplicated, or unsuitable for geographic analysis.

For example, an event with a missing location may still be useful for some aggregate statistics, but it cannot reliably be assigned to a geographic cell without a suitable location-reconstruction or correction process.

Typical data-quality checks include:

- Ensuring that the event type is recognized.
- Checking that latitude and longitude contain valid values when location is required.
- Ensuring that timestamps use a consistent representation.
- Validating numeric measurements and severity values where applicable.
- Identifying duplicate records when the available identifiers or source data permit it.
- Preserving the relationship between the imported event and its source.

These checks help prevent invalid records from distorting geographic risk scores.

The exact rejection, correction, and deduplication rules should follow the implemented ingestion logic. Normalization should not silently invent missing event locations or convert uncertain data into confirmed facts.

3.3. Step 3 — Associate events with stable geographic grid cells

One of the module's most important design principles is that historical events are aggregated into stable geographic grid cells, independently of OSRM road-segment identifiers.

This allows the platform to build persistent geographic risk profiles without making the identity of a risk area dependent on a particular routing-engine segment ID.

Road-segment identifiers may change because of changes in road-network data, map versions, or segmentation. A stable geographic grid provides a consistent spatial framework for aggregating historical events across imports and analysis periods.

How the grid-based approach works

1. The module obtains the geographic coordinates of a historical event.
2. It determines which grid cell contains the event's location.
3. It associates the event with that cell.
4. It aggregates the event with other relevant records assigned to the same cell.
5. It calculates or updates the cell's historical risk information.

Historical events

Events with available geographic coordinates

Stable geographic grid

Each event is assigned to the appropriate spatial cell

Per-cell event aggregation

Combine the relevant historical events and their weighted severity

Geographic risk profile

A normalized risk score associated with the cell

Why stable cells matter

Imagine that several speeding and hard-braking events are recorded at approximately the same location over multiple weeks.

If those events are aggregated only by a road-segment ID, changes to the routing engine's segmentation could make it harder to compare the records consistently.

With stable geographic cells, the platform can continue to associate those observations with the same spatial area, provided that the grid definition and coordinate-assignment rules remain consistent.

This approach supports longitudinal analysis: the platform can compare accumulated events in an area over time and identify places where elevated risk persists.

Important distinction: a geographic cell is not necessarily a single road segment. A cell may contain multiple roads, intersections, or road directions. Its risk score represents the events assigned to that geographic area, not necessarily the risk of every road within it.

The grid's dimensions and resolution influence the result. Smaller cells provide more spatial detail but may contain fewer events; larger cells aggregate more data but can blur differences between nearby roads. The exact cell size is an implementation parameter and should not be assumed without the relevant configuration.

3.4. Step 4 — Categorize events and determine their risk contribution

Not every historical event contributes equally to the risk assessment.

For example, speeding, hard braking, and harsh cornering represent different driving behaviours. Their implications also depend on factors such as severity, location, and the configured risk model.

The module uses weighted event severities to calculate geographic risk scores. Each relevant event contributes according to the applicable weighting rules rather than every event necessarily counting as an identical unit.

Conceptually, the process involves:

1. Identifying the event category.
2. Determining the applicable severity or weight.
3. Associating the weighted event with its geographic cell.
4. Combining its contribution with those of other relevant events.

5. Applying the risk-calculation rules to produce a normalized score.

An illustrative weighting table might look like this:

Event category	Illustrative risk consideration
Speeding	The degree or severity of the speeding event, when the required measurements are available.
Hard braking	The severity and frequency of hard-braking events.
Hard acceleration	The severity and frequency of hard-acceleration events.
Harsh cornering	The severity and frequency of harsh-cornering events.
Accident or other supported event	The applicable event category and weighting, if such records are included in the processed dataset.

This table describes the types of considerations a weighted model can use. It does not establish the exact numeric weights or prove that every category is currently active in the same production scoring formula.

Why weighting is necessary

Consider two geographic cells:

- Cell A contains several low-severity events.
- Cell B contains fewer events, but some have much higher assigned severity.

A simple event count might rank Cell A as riskier because it contains more events. A weighted model can account for severity and potentially identify Cell B as the higher-risk area.

The objective is to produce a score that reflects more than event volume alone.

However, event weighting and normalization must be designed carefully. A high score should be interpreted according to the model's definition, data coverage, and calibration—not as a direct probability that an accident will occur.

3.5. Step 5 — Calculate the normalized geographic risk score

After the relevant events have been associated with geographic cells and their severity weights determined, the module calculates a risk score for each applicable cell.

The score is normalized to the range:

$$0 \leq R_{cell} \leq 1$$

Where R_{cell} represents the historical risk score associated with a geographic cell.

A value closer to 0 represents a lower score under the configured model, while a value closer to 1 represents a higher score. These values should be interpreted according to the actual scoring rules and calibration.

A score of 0.8 does not automatically mean an 80% probability of an accident. It is a normalized risk indicator, unless the model has been specifically calibrated and validated to represent a probability.

Illustrative interpretation

Example of normalized cell scores

Illustrative values only; these are not official risk-classification thresholds.

Cell A

0.18

Lower relative score

Cell B

0.52

Intermediate relative score

Cell C

0.86

Higher relative score

The calculation may take into account the weighted severity and accumulation of events assigned to the cell. Depending on the implemented model, it may also use additional factors or normalization rules.

The exact mathematical formula, weights, treatment of event frequency, and handling of cells with insufficient data must be taken from the actual scoring implementation. They should not be inferred solely from the fact that the output is normalized between 0 and 1.

What the score provides

The score allows geographic areas to be compared using a common numerical scale.

It can support:

- Identifying areas with elevated historical driving-event risk.
- Comparing the relative risk scores of cells.
- Retrieving a risk indicator for a location.
- Combining cell-level information into a route-level assessment.
- Tracking changes in geographic risk when historical data is updated.

The score remains dependent on the quality and coverage of the underlying records. An area with few recorded events may have limited evidence, rather than necessarily being safe.

3.6. Step 6 — Build historical risk profiles

A geographic risk profile summarizes the historical event information associated with a particular area.

In addition to the normalized risk score, a profile can contain aggregate information such as event counts, average or maximum recorded speeds, and risk classifications, where those metrics are supported by the available data and implemented profile.

The purpose is to make the underlying history more interpretable than a single score alone.

Profile information	Purpose
Geographic cell identifier	Identifies the stable spatial area being analyzed.
Historical event counts	Indicates the number of relevant events accumulated in the area.
Event categories	Shows which types of driving behaviours contribute to the observed risk.
Weighted severity information	Helps explain the relative contribution of events to the score.
Normalized risk score	Provides the cell-level risk indicator between 0 and 1.
Average or maximum speed metrics	Provides additional context where speed measurements are available and included in the profile.
Historical time information	Helps interpret when the events occurred, if the profile retains temporal aggregation.

The precise profile fields depend on the available dataset and implementation.

Why a profile is more useful than a score alone

Two areas can have similar normalized scores but different underlying event patterns.

For example, one area may have accumulated many speeding events, while another may have a smaller number of severe braking events. The overall scores might be similar even though the underlying driving patterns differ.

The historical profile helps provide context for interpreting the score and identifying which event categories contribute to the observed risk.

It is also important to distinguish between a historical profile and a live traffic state. A historical profile summarizes previously recorded events; it does not necessarily indicate the current traffic situation, an active obstruction, or a collision occurring at that moment.

3.7. Step 7 — Point and radius risk queries

The module supports queries for a specific geographic point and for an area around that point.

The purpose is to retrieve historical risk information relevant to a location without requiring the client to retrieve and process the entire historical dataset.

A point-based query can be used when an application needs a risk indicator for a specific location. A radius-based query can retrieve information from the surrounding area.

The logical workflow is:

1. Receive the requested coordinates.
2. Determine the relevant geographic cell or cells.
3. Retrieve the corresponding historical risk profiles.
4. Apply the requested geographic scope, such as a radius where supported.
5. Return the relevant scores and associated information.

Example

A user selects a point on the map and requests historical risk information.

The module identifies the relevant geographic cell and retrieves its risk profile. If the request specifies a radius, the module can retrieve the applicable information from the surrounding area according to the query implementation.

Query type	Primary purpose	Typical result
Point query	Assess historical risk at a specific location.	The applicable cell's risk score and available profile information.
Radius query	Assess historical risk around a location.	Relevant nearby cells or aggregated information for the requested area.

The exact response structure, maximum radius, cell-selection logic, and aggregation rules depend on the implemented API.

A point query should not be interpreted as an exact risk assessment for an entire road segment unless the spatial resolution and association rules support that interpretation.

3.8. Step 8 — Route risk analysis

One of the module's most important applications is the assessment of historical risk along a journey.

Rather than evaluating a single point, a route query allows the platform to retrieve historical risk information associated with the geographic path of a route.

The route can cross multiple geographic cells, each with its own accumulated event history and normalized risk score.

The general process is:

1. Obtain the route geometry or the information needed to identify the route.
2. Determine which geographic cells are relevant to the route.
3. Retrieve the applicable cell-level risk profiles.

4. Combine or summarize those profiles according to the configured route-scoring logic.
5. Return the route-level risk information to the requesting application.

Requested route

A route represented by its geographic path

Identify relevant grid cells

Find the geographic areas crossed by or relevant to the route

Retrieve historical risk profiles

Collect the normalized risk scores and relevant historical information

Calculate the route-level assessment

Apply the configured aggregation and scoring rules

Return the route safety score

Provide the historical risk indicator to the requesting application

Why route analysis matters

Suppose two routes connect the same origin and destination.

One route may pass through areas with relatively low historical risk scores, while another crosses several areas with elevated scores. The module can provide information that helps compare the historical risk characteristics of the two routes.

However, a route safety score is not necessarily a recommendation to choose the route with the lowest score. Travel time, road accessibility, route length, current conditions, and the scoring methodology may also matter.

The exact route-level formula must be taken from the deployed implementation. It should not be assumed to be a simple average of the cell scores: the implementation may use other aggregation rules, and the same risk score can have different implications depending on how much of the route passes through each area.

The module provides historical risk information to support route assessment; it does not automatically establish that every route query triggers rerouting or navigation changes.

3.9. Step 9 — Produce a driver-facing route safety score

The module supports a driver-facing safety score based on the historical risk information associated with the requested route or geographic context.

Its purpose is to summarize a potentially large collection of historical event records into a result that is easier for a user to understand.

The score can help users compare routes or understand whether a journey crosses areas with elevated historical risk.

The important distinction is between:

- Cell risk score: a normalized historical risk indicator for a geographic grid cell.
- Route risk score: an assessment derived from the relevant geographic information along a route.
- Driver safety score: a user-facing score whose meaning depends on the implemented scoring definition.

These terms should not automatically be treated as interchangeable. A cell score measures the risk indicator for an area, while a route score summarizes multiple areas. A driver-facing safety score may present or transform the route result according to the application's design.

The exact score direction, display scale, formula, and interpretation must follow the implementation. For example, it should be clearly established whether a higher displayed score means safer or riskier before the value is presented to drivers.

A safety score should also be accompanied by an appropriate explanation of its historical basis. It should not be presented as a guarantee of safety or as a prediction that an accident will occur.

4. Historical data integration and the Ymane import

Historical data integration is important because the quality of the risk analysis depends on the amount, consistency, and relevance of the available event records.

The module supports batch and manual ingestion, but one particular integration needs to be distinguished from the general capability: the scheduled import from Ymane is currently suspended.

This means that the module can process available historical records, but it should not be assumed that new Ymane events are being imported automatically.

Aspect	Functional role	Current status
Batch import	Process historical event records in groups.	Supported by the module's ingestion workflow.
Manual import	Allow historical records to be supplied through the supported import process.	Supported.
Historical event processing	Normalize and associate available records with geographic cells.	Part of the historical risk workflow.
Ymane scheduled import	Periodically bring historical event data from Ymane into the platform.	Suspended.
Risk-profile updates	Update geographic risk information using records that have been successfully processed.	Depends on the data available and the configured processing workflow.

The distinction is operationally important: an ingestion capability does not guarantee that every intended source is currently supplying data.

If the Ymane integration is reactivated, its imported records would need to pass through the applicable data-quality, geographic-association, and risk-calculation stages before contributing to updated risk profiles.

5. Relationship with the other Road Risk Analysis modules

Module 4 complements the platform's real-time, AI-based, and community-driven capabilities by providing a historical perspective on road safety.

Related module	Relationship with Module 4
AI Hazard Detection — Module 1	Identifies visible road hazards from video. Historical risk analysis focuses on accumulated events and their geographic distribution. The two sources may complement each other where an appropriate integration exists.
Predictive Cooperative Collision Prevention — Module 2	Evaluates potential conflicts between connected vehicles using recent telemetry. Module 4 identifies longer-term patterns from historical records rather than predicting an immediate vehicle-to-vehicle conflict.
Community Reporting — Module 3	Collects user-submitted reports about road hazards and incidents. Those reports may provide additional historical context if they are integrated into the relevant risk dataset and scoring process.
Driver Behaviour Analytics — Module 5	Focuses on analyzing driver behaviour events. Module 4 uses accumulated historical events to establish geographic risk profiles and support point- or route-level assessments. The exact data-sharing relationship depends on the implemented integration.

Live Map	Can visualize geographic risk information if the relevant profiles and scores are exposed to the map interface. The map itself does not calculate historical risk.

The key distinction is that Module 4 focuses on where historical risk has accumulated and how that information can support geographic and route-level safety assessment.

6. Main inputs and outputs of the module

Category	Information
Primary inputs	Historical driving-event records, including speeding, hard braking, hard acceleration, harsh cornering, and other supported event categories.
Geographic inputs	Event coordinates and geographic grid definitions.
Supporting information	Event severity or measurements, timestamps, and source references where available.
Data-processing operations	Batch or manual ingestion, normalization, validation, geographic association, event aggregation, and weighted risk calculation.
Geographic aggregation output	Stable geographic cells containing aggregated historical event information.

Primary analytical output	Normalized cell-level risk scores ranging from 0 to 1.
Historical profile output	Geographic summaries and relevant event metrics where supported.
Query outputs	Point-based and radius-based risk information.
Route-analysis output	A route-level historical risk assessment based on the relevant geographic profiles.
Driver-facing output	A safety score or risk indicator according to the implemented scoring rules.

7. Important implementation distinctions and limitations

Capability	Interpretation
Historical event ingestion	Supports batch and manual imports; does not imply that every external source is active.
Ymane integration	The scheduled import is currently suspended.
Stable geographic grid	Provides a consistent spatial aggregation framework independent of OSRM road-segment IDs.

Weighted event severity	Allows events to contribute differently to risk according to the applicable weights.
Normalized risk score	Produces a value between 0 and 1; this is not automatically an accident probability.
Geographic risk profile	Summarizes accumulated historical events for a geographic area; it is not necessarily a live traffic state.
Point and radius queries	Retrieve relevant geographic risk information according to the implemented spatial rules.
Route risk analysis	Combines information relevant to a route according to the configured aggregation method.
Driver safety score	Must be interpreted according to its actual definition, scale, and scoring direction.
Data coverage	A low score or absence of recorded events does not necessarily mean that an area is safe; the underlying data may be incomplete.
Infrastructure decisions	Historical profiles can support investigations and prioritization, but should be interpreted alongside data quality and other relevant evidence.

8. Recommended improvements

The following are potential improvements to strengthen the reliability and usefulness of the module; they are not claims that these features are already implemented.

Improvement	Purpose	Expected benefit
Reactivate and monitor the Ymane import	Restore the scheduled ingestion workflow when operationally appropriate.	Improve the availability of historical records from that source.
Track data freshness and coverage	Record the period covered by the data and identify geographic areas with limited observations.	Reduce the risk of misinterpreting incomplete data as low risk.
Version risk-scoring parameters	Keep a record of the weights and normalization rules used for each calculation.	Improve reproducibility and explain changes in scores.
Explain route scores	Provide relevant information about the cells or event categories contributing to a route assessment.	Make route risk results more interpretable to users.
Validate score calibration	Compare the score's behaviour against suitable historical evidence and evaluation criteria.	Improve the reliability of the risk indicator.
Track historical changes	Compare geographic risk profiles across defined time periods.	Help identify persistent patterns and changes in observed risk.

One important priority is to make risk scores interpretable in the context of data coverage. An area with a low score and extensive historical observations should not necessarily be treated the same way as an area with a low score but very few recorded events.

9. Overall objective and expected result

The Historical Risk Analysis Module transforms accumulated driving-behaviour events into geographically organized risk information.

Its main value comes from combining:

1. Historical data ingestion to make available driving-event records usable for analysis.
2. Stable geographic aggregation to associate events with consistent spatial cells.
3. Weighted risk calculation to account for the relative severity of accumulated events.
4. Normalized geographic risk scores to provide a common scale for comparing areas.
5. Point and route queries to make historical risk information available to applications.
6. Driver-facing safety assessment to summarize route-related historical risk in a more accessible form.

The expected result is a historical risk layer that helps the platform identify areas where risky driving events have accumulated and provide useful information for route assessment and road-safety decision-making.

Module 5 — Driver Behaviour Analytics Module

1. Purpose of the module

The Driver Behaviour Analytics Module is responsible for analyzing driving-behaviour events to identify risky driving patterns, assess their geographical distribution, and provide information that can support driver safety monitoring and road-risk management.

Its primary objective is to transform individual driving events into structured behavioural indicators and geographic risk insights.

While Module 2 focuses on predicting potential collisions between connected vehicles, and Module 4 analyzes historical event patterns to calculate geographic risk scores, Module 5 focuses on the driving behaviours themselves: how drivers accelerate, brake, corner, and maintain their speed.

The module answers five fundamental questions:

- How are drivers behaving on the road? Identify events such as speeding, hard braking, hard acceleration, and harsh cornering.
- Where do risky driving events occur? Associate events with geographic locations to identify areas where these behaviours are frequently recorded.
- Which locations show recurring patterns? Aggregate events to identify geographic hotspots and risk zones.
- How can the results be interpreted? Provide behavioural statistics and visual representations such as heatmaps.
- How can the information support road safety? Help identify patterns that may require further investigation, driver coaching, or operational action.

The module therefore goes beyond simply recording individual events. It organizes behavioural data so that users can understand what types of risky driving occur, where they occur, and how frequently they are observed.

Main processing principle

Driving-behaviour event data

Speeding, hard braking, hard acceleration, harsh cornering, and other supported events

Event processing and categorization

Organize events by type, timestamp, location, and available severity information

Geographic association

Associate events with relevant road locations or geographic areas

Behavioural aggregation and analysis

Calculate event statistics and identify recurring patterns or concentrations

Heatmaps and risk-zone identification

Represent the geographical distribution of recorded driving behaviours

Safety insights and decision support

Support investigation, monitoring, and targeted road-safety actions

This diagram represents the logical workflow. The precise event sources, aggregation rules, and analytical outputs depend on the deployed implementation.

2. Scope of the module

The Driver Behaviour Analytics Module consists of capabilities for collecting or accessing driving events, organizing them, analyzing their distribution, and presenting the results through meaningful visualizations.

Component	Purpose	Expected result
Driving Event Collection	Obtain available driving-behaviour events from supported data sources.	Event records available for analysis.
Event Categorization	Organize events by behavioural category, such as speeding or hard braking.	Structured and filterable event records.
Event Severity Analysis	Use available measurements or severity classifications to distinguish event characteristics.	Additional context for interpreting driving events, where supported.

Geographic Event Association	Associate recorded events with their geographical locations.	Geolocated behavioural observations.
Behavioural Statistics	Aggregate events by category and other supported dimensions.	Statistics describing the frequency and distribution of recorded behaviours.
Geographic Heatmaps	Visualize concentrations of driving-behaviour events across a geographic area.	A spatial representation of observed event density or intensity.
Risk-Zone Identification	Identify geographic areas with elevated concentrations of relevant events according to the configured criteria.	Locations requiring further examination or road-safety attention.
Driver Behaviour Profiles	Organize behavioural information by driver or vehicle where the necessary identity data and functionality are available.	Behavioural summaries for supported entities.
Time-Based Analysis	Examine changes in recorded behaviours over time where timestamps and temporal queries are supported.	Insights into recurring patterns and changes in event frequency.
Analytical Visualization	Present event distributions and geographic patterns through dashboards, maps, charts, or other supported interfaces.	Information that users can interpret and use for decision-making.

The table describes the functional scope of driver behaviour analytics. Some capabilities, particularly individual driver profiles, temporal comparisons, and severity-based ranking, require confirmation against the actual implementation before being documented as active features.

3. Detailed functioning of the module

3.1. Step 1 — Collect driving-behaviour events

The process begins when driving-behaviour event records become available to the platform.

These events represent specific driving actions or conditions that have been detected, recorded, or supplied by a supported data source.

The principal categories identified for the platform include:

- Speeding.
- Hard braking.
- Hard acceleration.
- Harsh cornering.
- Other supported driving-behaviour events.

Each event represents an observation that can be analyzed individually or combined with other observations to identify patterns.

For example, a single hard-braking event may represent an isolated reaction to a traffic situation. Numerous hard-braking events concentrated in the same location may indicate a recurring pattern worth investigating.

The module's purpose is to make these events understandable and useful at both the individual-event and aggregate levels.

Typical event information

Field	Description	Analytical purpose

Event identifier	Unique reference for an event, where provided.	Supports event retrieval and traceability.
Event type	Behavioural category, such as speeding or hard braking.	Enables category-based analysis.
Event location	GPS coordinates or other available geographic reference.	Enables geographic analysis and mapping.
Event timestamp	Time associated with the recorded event.	Supports chronological ordering and time-based analysis.
Severity or measurement	Available information describing the intensity of the event.	Helps interpret the event where the relevant data exists.
Driver or vehicle reference	Association with a driver or vehicle, where available.	Supports entity-level analysis if implemented.
Source reference	Information identifying the originating data source, where recorded.	Supports traceability and data-quality investigations.

This is a conceptual list of useful event attributes, not a guarantee that every field exists in the current database.

The availability of driver identity, vehicle identity, measurements, and other attributes determines how detailed the analysis can be.

3.2. Step 2 — Categorize and prepare the events

Once event records are available, the module organizes them into consistent behavioural categories.

This is important because the platform needs to distinguish between different driving actions rather than treating every event as equivalent.

For example, speeding and hard braking describe different behaviours and may require different interpretations.

The preparation process can include:

1. Identifying the event category.
2. Checking that the relevant event information is usable.
3. Associating the event with its available location and timestamp.
4. Preserving severity or measurement information where provided.
5. Preparing the event for aggregation and visualization.

The precise validation and normalization rules depend on the implementation.

Example of event categorization

Event category	What it represents	What the analysis can reveal
Speeding	An event classified as driving above an applicable speed threshold.	Locations or periods where speeding events are frequently recorded.
Hard braking	A recorded braking event that meets the applicable hard-braking criteria.	Areas with repeated hard-braking observations that may merit investigation.
Hard acceleration	A recorded acceleration event that meets the applicable criteria.	Locations or patterns associated with repeated aggressive acceleration.

Harsh cornering	A recorded cornering event classified as harsh under the applicable rules.	Areas where harsh-cornering events are concentrated.

The exact thresholds used to classify these events must come from the event source or deployed configuration. They should not be assumed to be universal.

An event classification also does not necessarily establish the reason for the behaviour. Hard braking, for example, may be associated with unsafe driving, but it may also be a response to an unexpected obstacle or another road user.

3.3. Step 3 — Associate events with geographic locations

Geographic association is essential because the module aims to identify not only which behaviours occur, but also where they occur.

When an event includes valid GPS coordinates, the module can associate it with the relevant location for mapping and geographic aggregation.

This allows the platform to answer questions such as:

- Where are speeding events most frequently recorded?
- Which road areas have repeated hard-braking events?
- Where are harsh-cornering observations concentrated?
- Which geographic zones should be examined more closely?

The location-processing workflow can be represented as follows:

Driving event

An event record with available location information

Validate geographic information

Check whether the available coordinates or geographic reference can be used

Associate the event with an area

Use the relevant geographic representation for spatial analysis

Geographically organized event data

Events can be aggregated and visualized by location

The geographic representation may use coordinates, road references, or spatial areas depending on the specific processing component.

A location that cannot be reliably established should not be treated as an accurately geolocated event. The handling of events with missing or invalid coordinates depends on the implemented data-quality rules.

It is also important to distinguish between geographic event association and road-risk scoring. Module 5 organizes and analyzes driving behaviours geographically; Module 4 specifically provides historical risk analysis using stable grid cells and weighted event severities.

3.4. Step 4 — Aggregate driving-behaviour events

Individual events become more useful when they can be combined to reveal patterns.

The module can organize recorded events according to relevant dimensions, such as event category and geographic location. Where the necessary data and queries are supported, aggregation may also consider time periods, drivers, or vehicles.

The objective is to move from individual observations to a broader understanding of recorded driving behaviour.

For example, suppose a particular road area has accumulated numerous speeding events and hard-braking observations. Aggregating the events makes it possible to identify that these behaviours are concentrated in the same area.

Example of behavioural aggregation

The following figures are illustrative only and do not represent actual platform data.

Geographic area	Speeding events	Hard-braking events	Harsh-cornering events

Area A	18	7	3
Area B	5	16	4
Area C	11	6	14

These figures illustrate how different areas may exhibit different behavioural patterns.

- Area A has more recorded speeding events.
- Area B has more recorded hard-braking events.
- Area C has more recorded harsh-cornering events.

The purpose of the analysis is not simply to rank areas by total event count. Understanding the distribution by category helps distinguish the patterns that characterize each area.

A higher event count also needs to be interpreted in context. An area with substantially more traffic or more recorded driving time may naturally produce more events than a less-travelled area. Without appropriate exposure data, event counts alone cannot establish that the underlying driving behaviour is proportionally worse.

3.5. Step 5 — Generate behavioural heatmaps

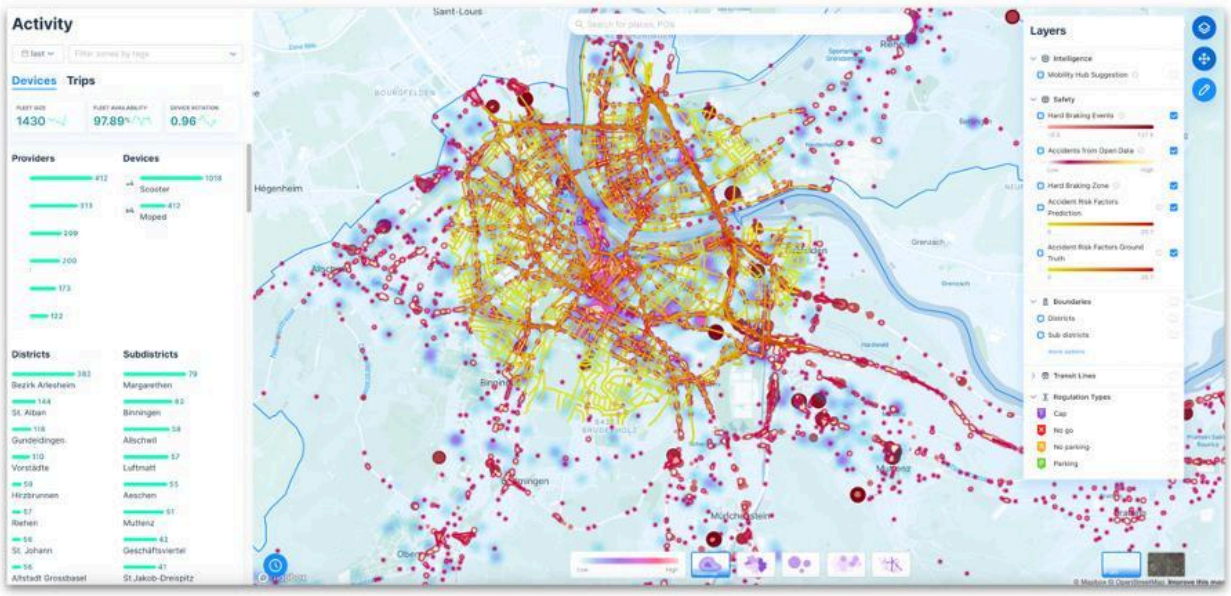
One of the principal analytical outputs of the Driver Behaviour Analytics Module is the geographic heatmap.

A heatmap represents the spatial concentration or intensity of recorded events. It allows users to identify areas where particular driving behaviours occur frequently.

Instead of examining every event individually, users can visualize patterns across a larger geographic area.

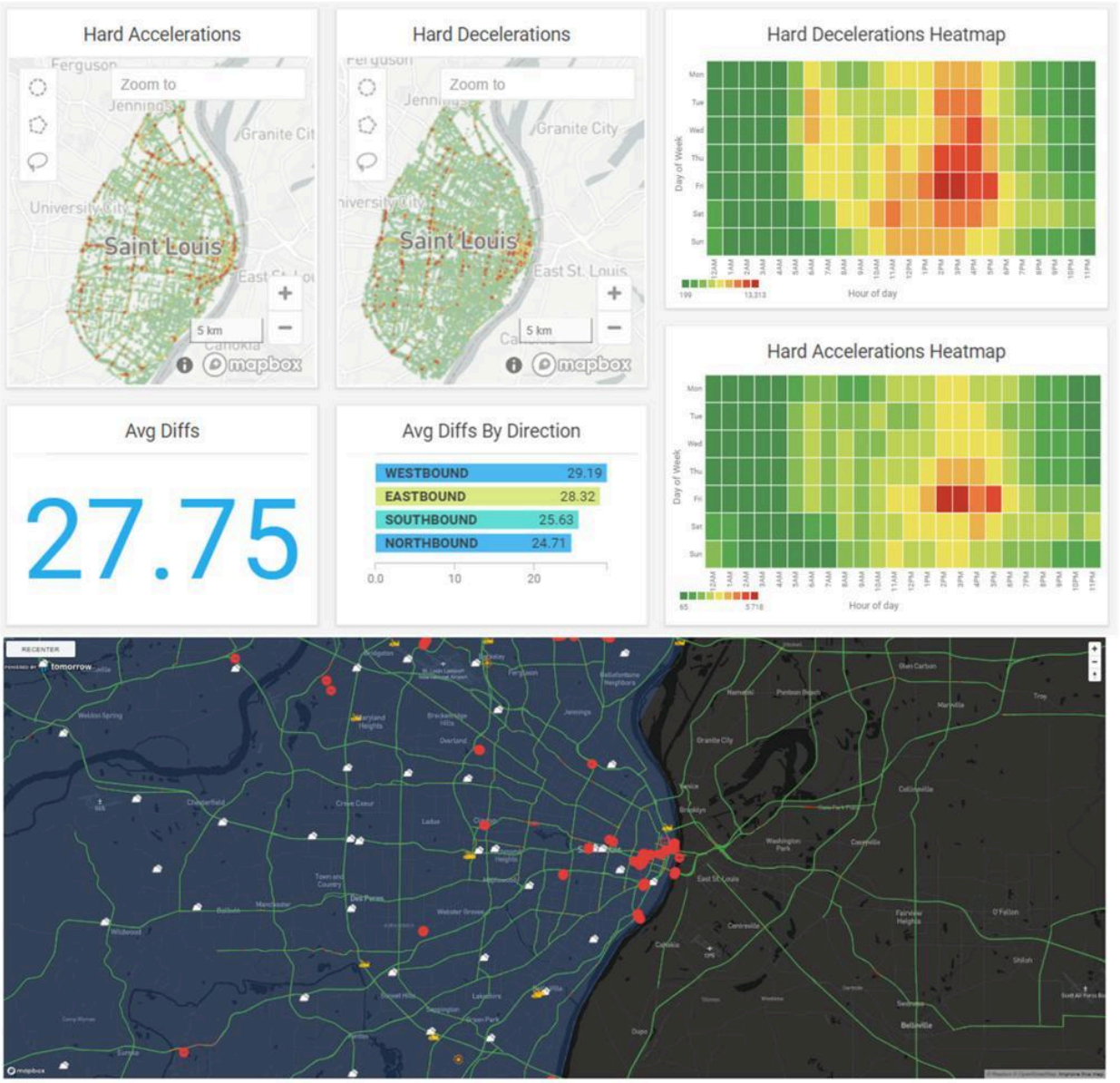
For example, a speeding heatmap can highlight where speeding events are concentrated, while a hard-braking heatmap can show areas with repeated hard-braking observations.

Conceptual heatmap views



Speeding heatmap

Shows where recorded speeding events are concentrated, helping analysts identify locations that may warrant closer examination.



Hard-braking heatmap

Highlights geographic concentrations of hard-braking observations that may be associated with traffic interactions, road conditions, or other factors.



Harsh-cornering heatmap

Shows where harsh-cornering events are concentrated and can help identify locations requiring additional investigation.

These are conceptual visual references, not screenshots of the deployed Road Risk Analysis application.

How a heatmap works

The general process is:

1. Select the relevant event category or categories.
2. Retrieve the applicable geolocated event records.
3. Aggregate the observations according to the configured spatial method.
4. Calculate the intensity or density represented by each geographic area.
5. Display the result on a map using a visual scale.

The exact spatial aggregation method and colour thresholds depend on the deployed visualization.

A heatmap's visual intensity should be interpreted according to its legend. It may represent event counts, event density, or another configured measure; it should not automatically be interpreted as a probability of an accident.

Heatmaps are useful for discovering patterns, but they do not independently explain why a behaviour occurs. A location with many hard-braking events may warrant investigation, yet the heatmap alone cannot determine whether the cause is congestion, a junction, road design, or another factor.

3.6. Step 6 — Identify geographic risk zones

In addition to visualizing individual event distributions, the module supports the identification of risk zones.

A risk zone is a geographic area identified as noteworthy according to the applicable behavioural-analysis criteria. For example, a zone may be highlighted because it contains a high concentration of speeding events or repeated observations of several risky driving behaviours.

The objective is to help analysts move from visual patterns to areas that deserve closer attention.

A conceptual workflow is:

Aggregated behavioural events

Event counts and geographic distributions by category

Apply the configured zone criteria

Evaluate the applicable event concentration or risk conditions

Identify relevant geographic areas

Determine which areas meet the implemented criteria

Present risk zones

Make identified areas available for visualization and further analysis

The exact criteria used to identify a risk zone—such as event-count thresholds, severity thresholds, density measures, or combinations of categories—must be taken from the actual implementation.

A risk zone should not be interpreted as a confirmed accident hotspot solely because it contains many behavioural events. It identifies a pattern that may deserve further investigation.

Potential applications include:

- Identifying areas with recurring speeding observations.
- Highlighting locations with repeated hard-braking events.
- Investigating road areas with concentrated harsh-cornering events.
- Supporting the prioritization of road-safety assessments.
- Comparing behavioural patterns between geographic areas.

These applications describe how the capability can be used. They do not imply that the module independently diagnoses the cause of every identified risk zone.

3.7. Step 7 — Analyze driver or vehicle behaviour where supported

Geographic analysis shows where events occur. Entity-level analysis, when supported by the available data, can help establish which driving behaviours are associated with a particular driver or vehicle.

For example, a driver or fleet manager may want to understand the frequency of recorded speeding events or hard braking over a defined period.

Such analysis requires reliable driver or vehicle identifiers associated with the event records. If the data does not include these identifiers, the module cannot reliably attribute the events to a specific driver.

A behavioural profile could contain information such as:

Information	Purpose
Event counts by category	Summarize the types of behaviours recorded.
Event timestamps	Support chronological analysis.

Event severity, where available	Distinguish events according to their recorded intensity.
Geographic distribution	Show where the events occurred.
Historical comparisons, where supported	Help identify changes in recorded behaviour over time.

These are potential profile elements rather than a guarantee that every one is available in the deployed application.

It is also important to distinguish an event-based profile from a validated driver-performance assessment. A high number of recorded events may reflect driving behaviour, exposure to traffic, operating conditions, or differences in the available data. Meaningful comparisons therefore require appropriate context.

3.8. Step 8 — Interpret behavioural patterns and support safety decisions

Once the module has categorized events, aggregated their distributions, and identified relevant geographic concentrations, its results can be used to support road-safety investigations.

Consider an area where repeated hard-braking events appear alongside a concentration of speeding observations.

This combination may indicate that the location warrants closer examination. Possible explanations could include traffic interactions, a junction, road geometry, or other conditions that influence driving behaviour.

The module provides evidence about the recorded pattern, but additional information is required to establish the cause.

Observed pattern	**Potential interpretation**	**Appropriate next step**

Repeated speeding events in one area	Speeding may be concentrated at that location.	Examine the event data and relevant road context.
High concentration of hard braking	Drivers may repeatedly be braking sharply in that area.	Investigate traffic interactions and road conditions.
Repeated harsh-cornering events	A particular road section or curve may warrant attention.	Examine the road geometry and supporting event records.
Several behavioural event categories concentrated in the same area	Multiple patterns may coexist in one geographic zone.	Investigate the combined evidence rather than relying on one metric alone.
Changes in event frequency over time	The observed pattern may be changing.	Compare periods using consistent data coverage and analytical criteria, where supported.

The intended result is to help analysts identify where additional attention may be useful, not to automatically attribute fault to drivers or determine the cause of a road-safety problem.

4. Relationship between Module 5 and Module 4

Modules 4 and 5 both analyze driving-related events, but they have different primary objectives.

Aspect	**Module 4 — Historical Risk Analysis**	**Module 5 — Driver Behaviour Analytics**

Primary objective	Calculate historical geographic risk indicators from accumulated event data.	Analyze driving behaviours and their geographic distribution.
Main analytical focus	Historical risk profiles associated with stable geographic grid cells.	Event categories, behavioural patterns, heatmaps, and risk zones.
Core processing	Weighted event aggregation and normalized risk-score calculation.	Behavioural categorization, aggregation, and geographic visualization.
Geographic representation	Stable grid cells independent of OSRM road-segment IDs.	Geographic association appropriate to the behavioural analysis and visualization.
Main outputs	Cell-level risk scores between 0 and 1, point and radius query results, and route risk assessments.	Behavioural statistics, heatmaps, and identified risk zones.
Driver or vehicle analysis	Depends on the historical risk data and scoring workflow.	Can support entity-level behavioural analysis if the necessary identifiers and functionality are available.
Typical use	Assess historical geographic risk and support route safety analysis.	Understand where and how risky driving behaviours are observed.

The modules can complement each other. For example, Module 5 may reveal that hard-braking events are concentrated in a particular area, while Module 4 can provide a historical risk score for the corresponding geographic cell.

However, the existence of both capabilities does not automatically establish that they use identical event datasets, the same aggregation formula, or a direct integration between their processing pipelines.

5. Relationship with the other Road Risk Analysis modules

Related module	Relationship with Module 5
AI Hazard Detection — Module 1	Identifies visible road hazards from video. These detections provide information about road conditions, whereas Module 5 analyzes driving behaviours.
Predictive Cooperative Collision Prevention — Module 2	Predicts potential conflicts between connected vehicles using recent telemetry. Module 5 analyzes recorded driving events rather than directly predicting vehicle-to-vehicle collisions.
Community Reporting — Module 3	Collects user-submitted road hazards and incident reports. These reports can provide complementary context for behavioural patterns if an appropriate integration exists.
Historical Risk Analysis — Module 4	Calculates historical geographic risk scores from accumulated event data. Module 5 provides a more behaviour-focused view through event categorization, heatmaps, and risk zones.
Live Map	Can visualize behavioural events or geographic risk zones when the corresponding data is exposed to the map interface. Visualization does not itself perform behavioural analysis.

The main distinction is that Module 5 focuses on understanding recorded driving behaviour and identifying where particular behaviours are concentrated, while the other modules focus on visual hazards, vehicle conflicts, community reports, or normalized historical risk scoring.

6. Main inputs and outputs of the module

Category	Information
Primary inputs	Recorded driving-behaviour events, including speeding, hard braking, hard acceleration, and harsh cornering.
Supporting inputs	Event timestamps, geographic coordinates, severity or measurement information, and driver or vehicle references where available.
Processing operations	Event categorization, geographic association, aggregation, statistical analysis, and identification of relevant event concentrations.
Primary analytical outputs	Behavioural event statistics and summaries by category.
Geographic outputs	Heatmaps showing event concentration or intensity according to the configured visualization.
Risk-zone outputs	Geographic areas identified according to the implemented behavioural-analysis criteria.

Optional entity-level outputs	Driver or vehicle behavioural summaries, where the required data and features are available.
Decision-support outputs	Information that can support further investigation, road-safety assessment, and targeted operational action.

7. Important implementation distinctions and limitations

Capability	Interpretation
Driving-event analysis	Organizes and analyzes supported behavioural events; event detection itself may depend on upstream data sources.
Speeding, hard braking, hard acceleration, and harsh cornering	These are the principal identified event categories. Their precise detection thresholds depend on the applicable configuration or source.
Geographic heatmaps	Show concentrations of recorded events; they do not automatically establish accident probability or causation.
Risk-zone identification	Highlights areas according to configured criteria; the exact thresholds and formulas must be verified against the implementation.

Driver-level analytics	Requires reliable driver identifiers and appropriate event associations. It should not be assumed to exist solely because event analytics are available.
Time-based comparisons	Require suitable timestamps, consistent data, and corresponding analytical functionality.
Severity analysis	Depends on the availability of event measurements or severity classifications.
Relationship with Module 4	The modules have complementary purposes, but identical datasets, formulas, or automatic data exchange should not be assumed.
Road-safety decisions	Analytical outputs can guide further investigation but do not independently prove why an event occurred or who was responsible.

8. Recommended improvements

The following are recommendations for strengthening the module. They are not claims that these features already exist.

Improvement	**Purpose**	**Expected benefit**

Normalize event rates by exposure	Consider relevant measures such as distance travelled or driving time when comparing event frequency, where the necessary data exists.	Make comparisons between areas or drivers more meaningful.
Add event severity context	Distinguish the intensity of recorded events where reliable measurements are available.	Improve interpretation beyond raw event counts.
Introduce temporal comparisons	Compare behavioural patterns across consistent time periods.	Help identify persistent or changing patterns.
Explain risk-zone criteria	Make the rules used to identify geographic hotspots transparent.	Improve interpretability and reproducibility.
Integrate relevant road context	Examine behavioural patterns alongside road geometry and other available information.	Support more informed investigations into possible contributing factors.
Provide traceable analytical results	Preserve relevant source information and the criteria used to generate analytical outputs.	Help users investigate and reproduce findings.

A particularly useful improvement would be to distinguish event concentration from event rate. An area with many recorded events may simply have a higher volume of traffic. Where reliable exposure data is available, event rates can provide a more meaningful basis for comparison.

9. Overall objective and expected result

The Driver Behaviour Analytics Module transforms recorded driving events into structured behavioural insights that can be interpreted geographically and operationally.

Its principal value comes from combining:

1. Event categorization to distinguish different driving behaviours.
2. Geographic association to establish where those behaviours are recorded.
3. Behavioural aggregation to identify recurring patterns and event concentrations.
4. Heatmaps to visualize the geographic distribution of recorded behaviours.
5. Risk-zone identification to highlight areas that may warrant further investigation.
6. Decision-support information to help users understand behavioural patterns and prioritize appropriate safety assessments.

The expected result is a behaviour-focused analytical capability that helps the Road Risk Analysis Platform identify where risky driving events are concentrated and provides evidence to support further investigation and road-safety action.